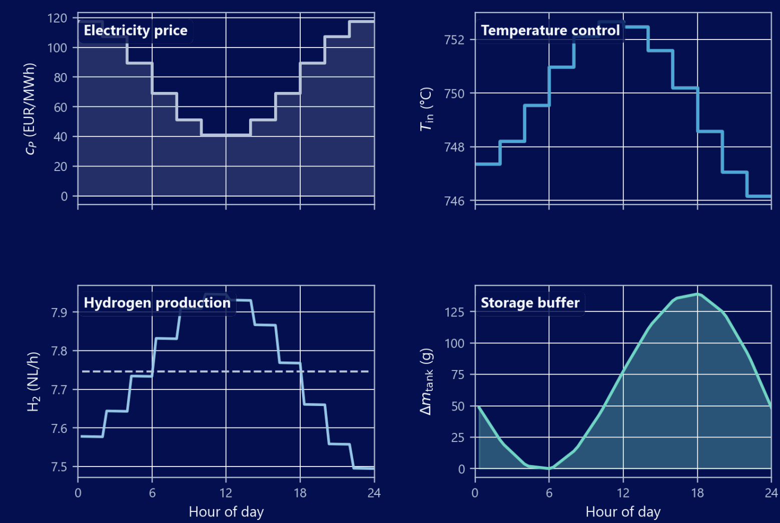
Optimal Control of SOEC's and sensitivity towards degradation

Analysis of the optimal control of SOEC's, where the modelling physical modelling of degradation is included, using finite volume method

Author:

Oliver Mohr (s214703)

s214703@dtu.dk
olmo@dynelectro.com



Supervisors:

DTU Energy:	Henrik Lund Frandsen (henlf@dtu.dk) Rafael Nogueira Nakashima (rafnn@dtu.dk)
DTU Compute:	Tobias Kasper Skovborg Ritschel (tobk@dtu.dk)
Dynelectro:	Thomas Erik Lyck Smitshuysen (thomas@dynelectro.dk)

Master Thesis
July 10, 2026

Contents

List of Figures

List of Tables

List of Symbols

Symbol	Unit	Description
Geometry, discretization, and time		
A_{cell}	m^2	Active electrode area
A_{int}	m^2	Flow-channel cross-sectional area
d	m	Cell thickness (channel height)
L	m	Channel length
w	m	Channel width
z	m	Axial coordinate along the channel
N	—	Number of axial channel segments
N_{cell}	—	Number of FVM control volumes along z
N_t	—	Number of control / time segments in the NLP
Δz	m	FVM mesh spacing
z_k	m	Centre of FVM cell k
Ω_k, V_k	m^3	FVM cell k and its volume
t	s	Time
T	s or days	Optimization horizon
Δt_k	s	Duration of control interval k
Species, flow, and thermodynamics		
$\mathcal{S}$	—	Gas species $\{H_2O, H_2, O_2\}$
$\mathcal{S}_\zeta$	—	Species on side $\zeta \in \{a, f\}$ (anode / fuel)
$\mathcal{M}$	—	Species index set for mass fractions y_k
c_i, μ_i	mol/m^3	Molar concentration; μ_i along channel (state)
x_i, y_k	—	Molar / mass fraction of species i or k
N_i, $N_{c,i}$	$\text{mol/(m}^2\cdot\text{s)}$	Molar flux; electrochemical source flux
$\dot{n}_i$, $\dot{V}_\zeta$	mol/s, $\text{m}^3\text{/s}$	Molar flow rate; volumetric flow on side ζ
v_{fuel}, v_{air}	m/s	Gas velocity on fuel / air side
p_i, P	Pa	Partial / total pressure
p_{ref}, p_{air}	Pa	Reference and air-side pressures
ρ	kg/m^3	Gas density
$c_{p,i}$, c_p, C_p	J/(mol·K), $\text{J/(K}\cdot\text{m}^3\text{)}$	Species, mixture, and volumetric heat capacity
$h_i(T)$, Δh	J/mol	Species enthalpy; reaction enthalpy change
Q_T	W/m^3	Volumetric heat source
Electrochemistry		
F	C/mol	Faraday constant

Symbol	**Unit**	**Description**
n_e	—	Electrons per reaction ($n_e = 2$)
ν_i	—	Stoichiometric coefficient of species i
i	$\mathrm{A/m^2}$	Current density
I, I_{cell}	A	Total cell current
U	V	Applied cell voltage (control)
U_{th}	V	Thermal (reversible) cell voltage
η_{cell}	V	Cell overpotential ($\eta_{\text{cell}} = U$)
η_{nernst}, η_{act}, η_{ohm}	V	Nernst, activation, and ohmic overpotentials
$\eta_{\text{act}}^{\zeta}$	V	Activation overpotential on side ζ
i_0^{ζ}	$\mathrm{A/m^2}$	Exchange current density on side ζ
R_{deg}, R_{ohm}	$\Omega{\cdot}\mathrm{m^2}$	Degradation and total ohmic resistance
R_{contact}	$\Omega{\cdot}\mathrm{m^2}$	Contact resistance
B_{ohm}	$\Omega{\cdot}\mathrm{m^2/K}$	Base factor in intrinsic ohmic term
$E_{\text{act}}^{\text{ohm}}$	J/mol	Activation energy in intrinsic ohmic term
ΔG	J/mol	Gibbs free energy change
Degradation		
$\mathcal{D}$	—	Set of slow degradation mechanisms
$\mathcal{D}_{\text{ohm}}$	—	Subset in R_{ohm} (Ni-mi, IC-ox)
$\mathbf{d}$, d_i	—	Degradation state vector and component i
$h_{\text{Ni-mi}}$, $h_{\text{IC-ox}}$	m	Ni-migration gap; interconnect oxide thickness
$s_{\text{IC-ox}}$	$\mathrm{m^2}$	Squared IC-oxide thickness (ODE state)
$d_{i,\text{critical}}$	m	End-of-life limit for mechanism i
$R_{\text{Ni-mi}}$, $R_{\text{IC-ox}}$	$\Omega{\cdot}\mathrm{m^2}$	Ohmic contributions from Ni-mi and IC-ox
σ_{elec}, $\sigma_{\text{elec},0}$	S/m	Electrode conductivity and pre-factor
$\sigma_{\text{IC-ox}}$	S/m	Oxide-layer conductivity
$E_{a,\text{elec}}$, $E_{a,\text{ox}}$, E_{el}	J/mol	Activation energies (electrode, IC ox., IC cond.)
$c_1, \ldots, c_4$	various	Fitted Ni-mi growth-rate coefficients
k_{mg}, k_{ox}	various	IC-oxidation rate and conductivity prefactors
ε, τ	—	Electrode porosity and tortuosity
Optimal control and economics		
$\mathbf{u}(t), \mathcal{U}$	—	Control vector (U, T_{in}); admissible set
T_{in}, T_{out}, T_{loss}	K	Inlet (control), outlet, and thermal recovery loss
$R(\mathbf{u}, \mathbf{d})$, R_{tank}	EUR/s	Revenue rate; contract rate with tank
P_{cell}, P_{heat}	W	Stack power; auxiliary sensible heating
φ, $\Psi(\mathbf{d})$, Ψ_{tank}	EUR	Integrated revenue; salvation; tank liquidation
c_{H_2}, c_P, $c_{P,k}$	EUR/kg, EUR/kWh	Hydrogen and power prices
$\bar{c}_P$, a	—	Mean diurnal power price and relative swing
c_{cell}	$\mathrm{EUR/m^2}$	Cell capital cost in Ψ
$\alpha(t)$, α_{Ψ}	—	Schematic weight; salvation exponent
κ_{buy}, κ_{sell}	—	Spot-purchase penalty; liquidation multiplier
m_{tank}, $n_{\text{purchase},k}$	mol, kg	Tank inventory; shortfall purchase
$\dot{n}_{H_2,\text{agreement}}$, $\dot{n}_{\text{prod}}$, $\dot{n}_{\text{purchase}}$	mol/s	Delivery, production, and purchase rates
$\Delta\dot{n}_{H_2}$	mol/s	Net H_2 production in R
PDE / FVM notation		

Symbol	**Unit**	**Description**
u, Q_u, F_u	various	Generic FVM field; source and flux terms
m_k	kg	Gas mass in FVM volume k
Greek symbols		
R	J/(mol·K)	Universal gas constant
α_ζ	—	Butler–Volmer symmetry factor; $\alpha_a = 0.65$, $\alpha_f = 0.59$
γ_ζ	A/m^2	Exchange-current pre-factor; $\gamma_a = 1.515{\times}10^8$, $\gamma_f = 7.666{\times}10^5$
E^ζ_{act}	J/mol	Electrode activation energy; $E^a_{\text{act}} = 1.40{\times}10^5$, $E^f_{\text{act}} = 1.05{\times}10^5$
m_i	—	Species exponent in i^ζ_0; $m_{\mathrm{H_2O}} = 0.33$, $m_{\mathrm{H_2}} = -0.1$, $m_{\mathrm{O_2}} = 0.22$
λ_a, λ_c	—	Exchange-current multipliers ($\lambda_a = 1$; $\lambda_c \approx 0.55$)
λ_T	W/(m·K)	Thermal conductivity (neglected)
Ω	m	Spatial domain
x_0	—	Reference molar-fraction scale ($p_{\text{ref}}/p_{\text{air}}$)

Greek parameter values follow Table 3.1 and Table 3.2 unless stated otherwise.

1 Abstract

This thesis studies the sensitivity of optimal control for a solid oxide electrolysis cell (SOEC) over its operating lifetime, with degradation included in the economic objective. A physics-based unit-cell model couples steady-state electrochemical and thermal fields along the gas channels (solved with a finite volume method) to differential degradation laws for nickel migration and interconnect oxidation, integrated forward in time with implicit Euler. On top of this forward model, an optimal control problem is formulated that maximizes integrated revenue (hydrogen sales minus electricity cost) plus a terminal *salvation* term penalizing proximity to end-of-life break limits. Cell voltage U and inlet temperature T_{in} are the primary controls; the resulting nonlinear program is solved by gradient-based optimization (SLSQP/Trust-Constr) using single-shooting forward replay.

Three result scenarios examine how much temporal control freedom matters and what drives the optimum. In an unrestricted 200-day baseline, the optimizer selects constant potentiostatic-isothermal (pot-iso) operation at an exothermal set-point ($U^* \approx 1.32\,\text{V}$, $T_{\text{in}}^* \approx 751\,^\circ\text{C}$); granting additional segment-wise freedom in U and T_{in} does not change the optimal trajectory. Nickel migration reaches the $5\,\mu$m break limit at $t = T$ and is life-limiting, while interconnect oxidation remains secondary. When a fixed hydrogen delivery contract and buffer tank are introduced, inventory-aware operation becomes relevant: the optimizer alternates gentle pot-iso periods with galvanostatic intervals dictated by tank inventory and penalty prices, at the cost of lower average net economy. A final 24-hour case with a diurnal power price and cyclic tank constraint shifts production toward low-price midday intervals mainly by modulating T_{in} while voltage stays essentially fixed.

IV validation, ODE replay against reference degradation simulations, and gradient checks support using the toolchain for qualitative optimal-control studies at engineering fidelity. The main conclusion is that optimal SOEC operation is scenario-dependent: ???

2 Introduction

This project aims to do the following:

The thesis studies the sensitivity of the optimal control for a solid oxide electrolysis cell (SOEC) over its lifetime, with the goal of minimizing degradation to prolong lifetime.

Starting by defining the optimal control problem. In loose terms, the objective maximizes revenue while penalizing degradation. A rough formulation is:

$$\max_u \left\{ \alpha(t) \int_0^T \text{revenue}\, \mathrm{d}t - (1-\alpha(t))\text{degredation} \right\} \tag{2.1}$$

The weighting function $\alpha(t)$ balances the importance of production versus degradation, and can be directly linked to real market prices. The details of how the overall optimization problem is formulated are provided later; at this stage, this serves as a broad overview. The core focus is to examine the optimal control $u^*(t)$ (the set of control choices that maximize the objective) and to see what happens as we give the controller more freedom, and lesser fixation. Before diving into the technical setup, the basic idea of giving more freedom will be demonstrated using a simple example.

2.1 A Simple Example of Control Freedom

Consider a simple example: a water gun at a fixed spot that shoots with constant muzzle speed v_0. The only thing we can control is the angle α. The angle sets how far the water goes; the maximum range is at 45°, so restrict α to $[0^\circ, 45^\circ]$ and call the range at 45° simply L_{max} [20].

The setup is illustrated below:

Thomas (a co-supervisor) stands somewhere on the field at L_T. The goal is to hit him, or get as close as possible. In symbols, something like $\min_\alpha |L(\alpha) - L_T|$.

If L_T is within range, pick the angle that lands on him. If he is too far away, shoot at 45° and accept the miss. The optimal angle as a function of where Thomas stands can be sketched as in Figure 2.2.

Figure 2.2: Optimal angle when only α can be chosen and $v = v_0$ is fixed.

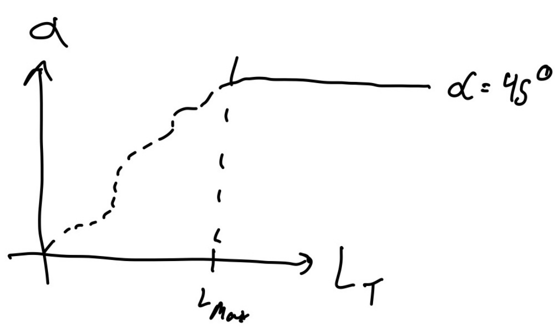


Figure 2.3: Optimal aim when v is fixed (angle varies) versus when v can also be chosen (angle stays at 45°, speed varies).

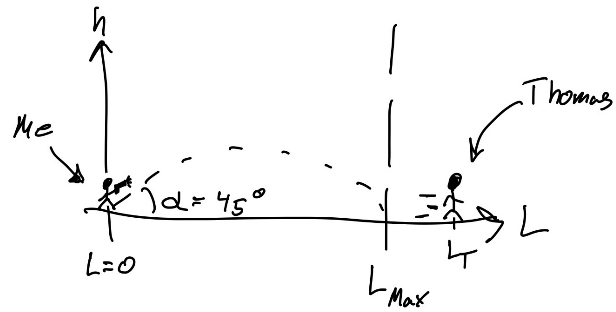


Figure 2.1: Sketch of the optimization problem: the aim is to hit as close to Thomas as possible.

So far so obvious. The point of the example is what happens when we add another knob.

Suppose we can also choose the muzzle speed v, not only the angle. Then the picture flips: instead of carefully tuning α for every L_T, it is natural to leave the gun at 45° (maximum range for a given speed) and adjust v until the water reaches Thomas. Targets that were out of range before can be reached by turning up the pressure; targets that were already in range can still be hit, but the optimizer no longer needs to lean on the angle the way it did when v was fixed.

Figure 2.3 sketches the contrast: one curve for the old case (angle does the work), one for the new case (angle stays near 45°, speed does the work).

The takeaway is loose but important: giving the optimizer more freedom does not just add options — it can change what the first control variable is asked to do.

The same idea carries over to the cell problem in Equation 2.1. If only the cell voltage U is free and the inlet temperature T_{in} is fixed, the optimizer works mainly through U. Once T_{in} is also a decision variable (as in Equation 4.1), the optimal voltage profile can look quite different — just as the water gun stops fiddling with the angle once the pump pressure is on the table. The full setup is spelled out later in the optimal-control chapter.

3 Physical Modelling

3.1 Modelling govering SOEC equations

3.1.1 Introduction

In this chapter the aspects of solid oxide electrolysis cell's (SEOC's) will be introduced. Briefly visiting different scales of operation, before a more detailed description of how the mole flow is modelled, rounding it up with the energy balance that determines the behavior of temperature flow.

The basics are presented so that readers with a background in applied mathematics can follow the system. Derivations throughout the chapter therefore assume familiarity with standard intermediate steps; lengthier or less routine derivations are given in the main text or in an appendix. Readers who wish to proceed directly to the governing equations will find them summarized in Table 3.1.

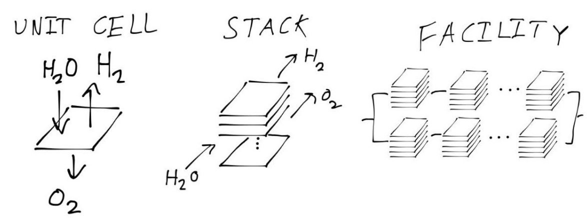


Figure 3.1: A sketch of the scales: the three different scales mentioned in this paper from smallest (left) to largest (right)

The three different scales of SOEC discussed in this paper are defined as follows. The first and smallest scale is the unit cell, where the focus is only on the dynamics occurring within a single SOEC cell. We will spend the majority of the Thesis on this scale.

Next is the stack. At this level, multiple unit cells are stacked together, which naturally introduces more complexity to the system. Figure 3.1 is a quick visualization of the different scales.

Finally, the largest scale considered is at facility scale. At this level, multiple SOEC stacks are connected together; the facility operates much like an electrical circuit, with each stack behaving as its own component with distinct characteristics. However as stated, we will be focusing on the cell scale. And might be relating this to the operation of other scales.

To draw a parallel with the earlier water gun example: in this chapter, we begin constructing the "water gun" of our model, step by step, laying out its fundamental mechanics and underlying principles.

3.1.2 Basics of SOEC

The “water gun” also known as the unit cell, can in simple terms be described by Figure 3.2, here it is visible what happens at the small scales.

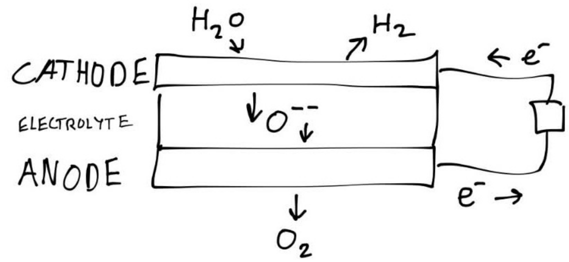


Figure 3.2: A sketch of a unit cell: the basic principles of an SOEC cell are illustrated. The cathode pulls apart the hydrogen and oxygen in the water. The oxygen is transferred through the electrolyte, then formed to dioxygen in the anode, all due to an applied current.

The unit cell is composed of three main components: the **cathode**, the **electrolyte**, and the **anode**. At the cathode, water molecules are split into dihydrogen (H_2) and oxide ions (O^{--}). This process is only possible due to the presence of electrons provided by an external electric current: these electrons participate directly in the reaction at the cathode, facilitating the reduction of water.

The newly-formed oxide ions then migrate through the solid **electrolyte**, propelled by the electric field generated by the applied voltage, highlighting the central role of electrons and electrical energy in driving the overall process. Once the oxide ions reach the **anode**, they release their excess electrons and recombine to form dioxygen (O_2). The electrons that are released here travel through the external circuit, closing the electrical loop.

Summing up, the operation of the SOEC is fundamentally tied to the movement of electrons, both in the chemical reactions at the electrodes and in sustaining the ionic and electronic currents through the device. This is captured in the overall reaction, $2H_2O + 4e^- \rightarrow 2H_2 + 2O^{--} \rightarrow 2H_2 + O_2 + 4e^-$, where the electrons facilitate water splitting at the cathode and are recovered at the anode. For modeling purposes, and since the focus will generally not be on the internal mechanisms, the model simplifies to the absolute global equation (eq. 3.1).

$$H_2O \rightarrow 2H_2 + O_2 \tag{3.1}$$

To clarify some terminology. The upper side of the unit cell—where water is supplied and dihydrogen (H_2) is produced—is called the **fuel side**. This is because water acts as the “fuel” for the process, supplying the reactant that gets split. The lower side—where dioxygen (O_2) is released—is called the **air side**, since atmospheric air is usually introduced here as the inlet. These terms, “fuel side” and “air side,” are commonly used throughout SOEC literature to describe the two compartments of the cell.

3.1.2.1 The chemical flow in a unit cell

The equation in 3.1 treats the reaction as strictly one-way, but in reality, some produced hydrogen and oxygen can recombine to form water again, especially as the system approaches equilibrium based on its electrical state. This reverse reaction is rare at the start but becomes more significant near equilibrium. Yielding equation 3.2.

$$2H_2O \underset{p_r}{\overset{p_l}{\leftrightarrows}} 2H_2 + O_2 \tag{3.2}$$

Here, p_l stands for the transition probability that two hydrogen and one oxygen molecule will combine and turn back into water (going from right state $\rightarrow$ left state), while p_r is the probability for splitting water into hydrogen and oxygen (left state $\rightarrow$ right state).

From these probabilities, the production rate of H_2O at every point in the cell can be determined (assuming concentrations and probabilities are known everywhere). The difference between how much H_2O is produced versus how much is consumed is taken. Keeping in mind that this is calculated per unit volume of the cell, it has the unit $\frac{\mathrm{mol}}{\mathrm{s \cdot m^3}}$:

$$R_{\mathrm{H_2O}} = p_r c_{\mathrm{H_2O}}^2 - p_l c_{\mathrm{H_2}}^2 c_{\mathrm{O_2}}$$

Where c_x is the concentration $\left(\frac{\mathrm{mol}}{\mathrm{s \cdot m^3}}\right)$ of molycule x

To obtain the production rate for the whole cell (the global rate), this quantity is integrated over the thickness (height) of the cell. Which gives the total rate $N_{c,\mathrm{H_2O}}$ (later this will be treated as a local source, but for now it is kept global):

$$N_{c,\mathrm{H_2O}} = \int_0^d R_{\mathrm{H_2O}}$$

This gives the average production rate per area of the cell, which is a useful parameter to work with. But realistically, calculating this from scratch could get a bit complicated. Thankfully, Faraday already did the heavy lifting here. By Faraday's law [18], the rate per cell area can be written as in equation 3.3:

$$N_{c,\mathrm{H_2O}} = \frac{i}{2F} \tag{3.3}$$

Where i is the current, F is Faraday's constant, and the 2 comes from the number of electrons involved in the chemical equation.

For now, equation 3.3 is kept as it is, since the formulation is still at the global level; hence the current in the equation is the global current. Later, when making all this local, the current must be computed using the Nernst equation. More about that later. First, a form that describes the system locally is introduced.

3.1.2.2 Behavior through a channel over the cell

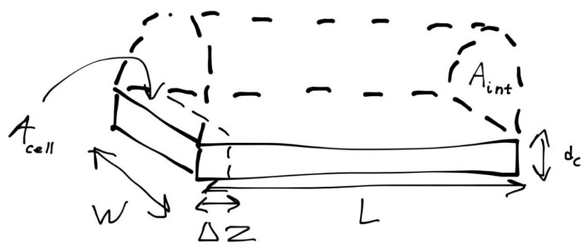


Figure 3.3: Schematic illustration of channel discretization for modeling local behavior along the fuel side of a SOEC. The channel is divided into N segments of length Δz, showing the flow of steam over the cathode.

To describe the system locally, figure 3.3 is used. The fuel side is modeled as a channel (the upper section in the figure), where the steam (H_2O) flows over the cathode. The width of the channel is w and the length is L, with the channel extending from the inlet at $z = 0$ to the outlet at $z = L$ (where L is the length of the cell). The gas flows through a constant cross-sectional area A_{int}.

For any control volume within this channel, mass conservation dictates that the accumulation of water over time equals the flow in plus the production of the cell reaction minus the flow out. Following the formulation by Primdahl and Mogensen [14] for a continuously stirred tank reactor (CSTR), the mass balance for the water molar fraction $x_{\mathrm{H_2O}}$ is:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = A_{int}N_{in,\mathrm{H_2O}} - A_{int}N_{out,\mathrm{H_2O}} + A_{cell}N_{c,\mathrm{H_2O}} \tag{3.4}$$

Where V is the volume of the gas in the reactor, N_i is the molar flux for species i, and A_{cell} is the active electrode area.

To obtain the local behavior along the channel, discretization along the channel length is required, meaning the length is split into sections. This is also illustrated in figure 3.3. Hence $z_i - z_{i-1} = \Delta z$ for all $i = 1, \ldots, N$, with $z_0 = 0$ and $z_N = L$, giving a sequence:

$$z_0 < z_1 < \cdots < z_N$$

Each section $z \in (z_{i-1}, z_i)$ can then be viewed as its own cell, where the inlet flow at z_{i-1} is the outlet flow of the previous interval. Letting $\Delta z \to 0$ and inserting the definition of $N_{c,\mathrm{H_2O}}$ (equation 3.3), and the PDE in equation 3.5 pops right out. (Appendix A1)

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}N_{c,\mathrm{H_2O}} - \frac{\partial N_{\mathrm{H_2O}}}{\partial z} \tag{3.5}$$

Where $N_{\mathrm{H_2O}} = v_{fuel}\,\mu_{\mathrm{H_2O}}$ and $N_{c,\mathrm{H_2O}} = \nu_{H_2O}\frac{i}{n_e F}$, whith n_e the number of electrones in the reaction being 2.

By similar computations for H_2 and O_2 the following equations will appear:

$$\frac{\partial \mu_{\mathrm{H_2}}}{\partial t} = \frac{w}{A_{int}}N_{c,\mathrm{H_2}} - \frac{\partial N_{\mathrm{H_2}}}{\partial z} \tag{3.6}$$

$$\frac{\partial \mu_{O_2}}{\partial t} = \frac{w}{A_{int}} N_{c,O_2} - \frac{\partial N_{O_2}}{\partial z} \tag{3.7}$$

Where $N_{H_2} = v_{fuel}\,\mu_{H_2}$ and $N_{O_2} = v_{air}\,\mu_{O_2}$, with ν_i the stochiometric coefficient for species i, μ_i the molar concentration of species i, and $v_{fuel/air}$ the corresponding velocity. And $N_{c,i} = \nu_i \frac{i}{n_e F}$.

From the chemical equation of the reaction:

$$2H_2O \leftrightharpoons 2H_2 + O_2$$

The stoichiometric coefficients are determined as $\nu_{H_2O} = -\nu_{H_2} = -2\nu_{O_2} = 1$.

This equation describes the local evolution of the species concentration. As noted earlier, the current i is no longer a global current. To determine it, the Nernst equation and more is required; this will be explained shortly.

3.1.2.3 Introducing the overpotentials

From the equation stated in 3.5 only the current density i remains to be determined. The control variable is the voltage U, which equals the overpotential of the whole cell (η_{cell}) or cell voltage.

The cell voltage can be written as the sum of all overpotentials over the cell as in equation 3.8. Concentration overpotentials that appear when accounting for diffusion are neglected here, and the concentration is assumed to be the same everywhere on each side of the cell. (Section 3.3)

$$\eta_{cell} = \eta_{nernst} + \eta_{act} + \eta_{ohm} \tag{3.8}$$

$$\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log\left(\frac{x_{H_2}\sqrt{x_{O_2}}}{x_{H_2O}\sqrt{x_0}}\right), \tag{3.9}$$

Where $x_0 = p_{ref}/p_{air} \approx 1$, ΔG is Gibbs free energy approximated by the polynomial $\Delta G = -0.0031\,T^2 - 49.119\,T + 244778$. [4]

$$\eta_{ohm} = iR_{\text{deg}} \tag{3.10}$$

Where R_{deg} is the resistince induce from degradation mechanisms.

$$\eta_{act} = \eta_{act}^a + \eta_{act}^c \tag{3.11}$$

Where η_{nernst} is the Nernst overpotential, representing the voltage generated by the electrochemical reaction, and η_{act} is the activation overpotential, which accounts for the energy barrier that must be overcome for the reaction to proceed, η_{ohm} is the Ohmic resistence, which accounts for resistance that is active due to degredative mechanisms. [4]

The Ohmic overpotential is essentially Ohm's law with some resistance; but it is in this resistance we get the degradation, or in terms of the water gun problem, we find where Thomas is standing within the ohmic resistance. The Nernst overpotential describes the willingness of the reaction to proceed. When there is a lot of H_2O and very little H_2 and O_2, the Nernst overpotential is large in absolute value but negative. This drives the reaction away from the high concentration in H_2O and closer to equilibrium. Away from the Gibbs term and the other overpotentials, the *equilibrium* occurs where the fraction is 1. The Nernst overpotential can therefore be viewed as the *force* that pulls the concentrations towards equilibrium.

The activation overpotential is probably the most difficult part to compute. The governing equations are listed first; their meaning is explained afterwards.

The overpotentials for the anode and cathode arise by solving Equation 3.12, also known as the Butler-Volmer equation.

$$i = i_0^\zeta \left(\exp\left[\alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta\right] - \exp\left[-(1-\alpha_\zeta)\frac{n_e F}{RT}\eta_{act}^\zeta\right] \right), \qquad \zeta \in \{a, f\}. \tag{3.12}$$

Where α_ζ is the symmetry factor for the corresponding side, and i_0^ζ is the exchange current for the respective side, with $\alpha_a = 0.65$ and $\alpha_c = 0.59$. [13]

The exchange current is computed as in Equation 3.13.

$$i_0^\zeta = \lambda_\zeta \gamma_\zeta \exp\left[-\frac{E_{act}^\zeta}{RT}\right] \prod_{i \in \text{Species}_\zeta} \left(\frac{x_i}{x_0}\right)^{m_i}, \qquad \zeta \in \{a, f\}. \tag{3.13}$$

Where Species$_a$ $= \{O_2\}$ and Species$_c$ $= \{H_2O, H_2\}$. $\lambda_a = 1$ (in principle it does not appear in the equation) and λ_c is also determined by degradation mechanisms, discussed later. γ_ζ is the pre-exponential factor for the respective side with values $\gamma_a = 1.515 \cdot 10^8$ and $\gamma_f = 7.666 \cdot 10^5$. E_{act}^ζ is the activation energy, with $E_{act}^a = 1.40 \cdot 10^5$ and $E_{act}^f = 1.05 \cdot 10^5$. And $m_{H_2O} = 0.33$, $m_{H_2} = -0.1$ and $m_{O_2} = 0.22$. [13]

This describes the electrical energy required to overcome the energy barrier for the reaction. Somewhat similar to how water does not necessarily freeze at 0 C°. There is some discussion in the field about whether the Butler-Volmer equations are adequate for this phenomenon; but for this project the Butler-Volmer model is taken as a sufficient approximation.

3.1.2.4 Computing the current i

With the overpotentials introduced, it is clear that some of them depend on the current i, which is the missing piece in 3.3 needed to complete the coupled differential equations 3.5, 3.6 and 3.7.

The current is obtained by solving Equation 3.8, with $\eta_{cell} = U$, and i defined as in Equation 3.14

$$i = \underset{i}{\text{Solve}}\left[U - \eta_{nernst} = \eta_{ohm}(i) + \eta_{act}(i)\right] \tag{3.14}$$

Using various simplifications this equation can be solved in different ways; for now it is solved numerically, using a root solver from the python library Scipy [19].

With the equations for mole transport in place, the temperature behavior is treated next, before all equations are summarized in a table.

3.1.3 Energy Balance

To derive the energy balances, the derivation starts at the thermal energy equation given by Equation 3.15 [11]. A number of simplifications are then applied to obtain Equation 3.16.

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T \tag{3.15}$$

where ρ is density, h specific enthalpy, λ_T thermal conductivity, h_k and N_k the specific enthalpy and flux of species k, and Q_T the volumetric heat source (e.g. electrochemical heating).

Starting with Equation 3.15, essentially stating how much energy a volume accumulates, in enthalpy per unit volume (left-hand side). Is equal to the compression (The change in pressure) plus the energy added by heat conduction (Second order differential term in temperature) and the transportation of enthalpy by species (H_2O, H_2, O_2), and lastly adding the reaction heat Φ.

The first simplifications is to neglect the energy accumulation due to heat conduction ($\nabla \cdot (\lambda \nabla T) = 0$). Also assuming constant pressure, $h \approx c_p T$ and $\partial_t p = 0$, which yields:

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + Q_T \tag{3.16}$$

Where $c_p = \sum_{e \in \mathcal{M}} c_{p,k} y_k$, with y_k being the mass fractions (Not the molar fraction), for further information look at Appendix A2

It remains to define the source term, which follows from the power expression in Equation 3.17. Power is the energy per unit time, hence the energy added to the system.

$$Q_T = \frac{i\, w}{A_{int}} U \tag{3.17}$$

3.1.3.1 At steady state

At steady state the expression for the molar progression can be used to simplify the equations further, yielding Equation 3.18.

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z} \tag{3.18}$$

Where $U_{th} = \frac{\Delta h}{n_e F}$ with $\Delta h = h_{\mathrm{H_2O}} - h_{\mathrm{H_2}} - \frac{h_{\mathrm{O_2}}}{2}$, (Appendix A2).

With the energy balance, or temperature differential equation, in place, the cell model is complete. The following section takes a deep dive into “Thomas’s position” and “how he moves” also known as the degradation mechanisms.

Description	Equation	Ref.
Faraday's law	$N_{c,i} = \nu_i \frac{i}{2F}$	Equation 3.3
Molar concentration DEs	$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}} N_{c,i} - \frac{\partial N_i}{\partial z}$	Eqs. 3.5, 3.6, 3.7
Cell voltage	$\eta_{cell} = \eta_{nernst} + \eta_{ohm} + \eta^a_{act} + \eta^c_{act}$	Equation 3.8
Ohmic overpotential	$\eta_{ohm} = iR_{deg}$	Equation 3.10
Nernst overpotential	$\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log \left(\frac{x_{\mathrm{H_2}} \sqrt{x_{\mathrm{O_2}}}}{x_{\mathrm{H_2O}} \sqrt{x_0}} \right)$	Equation 3.9
Butler–Volmer	$i = i_0^{a/f} \big(\exp[\alpha_{a/f} \frac{n_e F}{RT} \eta^{a/f}_{act}] - \exp[-(1-\alpha_{a/f}) \frac{n_e F}{RT} \eta^{a/f}_{act}] \big)$	Equation 3.12
Exchange current	$i_0^{a/f} = \lambda_{a/f} \gamma_{a/f} T \exp\big[-\frac{E^{a/f}_{act}}{RT} \big] \prod_{i \in \mathcal{S}_{a/f}} (x_i/x_0)^{m_i}$	Equation 3.13
Gibbs free energy	$\Delta G \approx -0.0031T^2 - 49.119T + 244778$	
Energy balance	$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + \frac{i\, w}{A_{int}} U$	Eqs. 3.16, 3.17

Table 3.1: Overview of model equations (Here without degradation which is in Section 3.2.1).

3.2 Modelling degredation

3.2.1 Introduction

This section examines the two components R_{ohm} that appears in Equation 3.10 and λ_c which appears in Equation 3.13. The latter is more complicated, so the Ohmic resistance is treated first. Below a equation is set up, which describes how the degradation plays a role on the ohmic resistance.

$$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left\{\frac{E_{act}^{\text{ohm}}}{RT}\right\} + R_{contact} + \sum_{d \in \mathcal{D}_{\text{ohm}}} R_d$$

Here, the first two terms are time-invariant and depend only on the present conditions of the cell. In contrast, the degradation terms, represented by the summation, evolve and increase as the cell ages. The first degradation mechanism considered is the resistance arising from nickel migration.

3.2.2 Nickel migration

Nickel migration is a simple phenomenon to understand. Essentially the electrode is saturated with nickel, but as time goes by the saturated part of the cell will shrink, hence we will have a gab in the electrode where there are no nickel. Since the nickel plays a huge part in transporting the oxygen ions fast and smoothly, the absence of nickel will induce an added travel resistance for the ions. Now the resistance can be represented by Equation 3.19.

$$R_{\text{Ni-mi}} = \frac{h_{\text{Ni-mi}}}{(\frac{\varepsilon}{\tau}\sigma_{\text{elec}})} \tag{3.19}$$

Where $d_{\text{Ni-mi}}$ is the thickness of nickel absence layer, which develops over time, σ_{elec} is the electrode conductivity, ϵ is the zirconia fraction and τ is the tortuosity of the zirconia fraction [4]

The sketch in Figure 3.4 illustrates how the nickel migration develops over time, and the impact it has.

The conductivity of the electrode is given by Equation 3.20. This conductivity is in reality the reduction in conductivity due to the creep of the nickel part of the electrode.

$$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,0}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right) \tag{3.20}$$

Javid Beyrami et al. [4] describe the development of nickel migration by the following differential equation, where c_1 and c_2 are negative and c_3 and c_4 are positive. For high values of current and temperature, migration increases; if either T or i is too small or too large, migration also increases.

$$\frac{\partial h_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_0$$

Where c_k is fitted to experimental data.

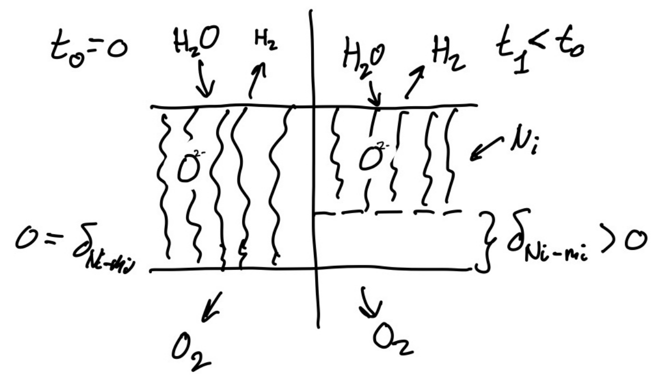


Figure 3.4: For a cell without degradation (Left), nickel migration has not occurred and $d_{\mathrm{Ni-mi}} = 0$ at the layer between the nickel-filled electrode and the electrolyte. Over time the nickel-filled part of the electrode creeps away from the electrolyte (Right), giving rise to resistance induced by the raw electrode on the traveling species.

3.2.3 Interconnect Oxidation

Another important degradation mechanism to consider is interconnect oxidation. This process also contributes to the overall ohmic resistance, and is relatively straightforward to understand: it occurs when the metallic interconnect undergoes oxidation, in other words, when the metal effectively "rusts."

The resistance resulting from interconnect oxidation is in structure much like that of nickel migration. Specifically, it is determined by the thickness of the oxidized layer divided by its (reduced) electrical conductivity, as shown in Equation 3.21 [4], [12], [5].

$$R_{\mathrm{IC-ox}} = \frac{h_{\mathrm{IC-ox}}}{\sigma_{\mathrm{IC-ox}}} \tag{3.21}$$

Where the oxidized interconnect conductivity is given as a function of temperature in Equation 3.22.

$$\sigma_{\mathrm{IC-ox}} = \frac{k_{\mathrm{ox}}}{T} \exp\left\{-\frac{E_{\mathrm{el}}}{RT}\right\} \tag{3.22}$$

Beyrami et al. [4] describe the growth of the oxide layer by the following rate law.

$$\frac{dh^2_{\mathrm{IC-ox}}}{dt} = k_{\mathrm{mg}} \exp\left\{-\frac{E_{a,\mathrm{ox}}}{RT}\right\} \tag{3.23}$$

The squared-thickness form in Equation 3.23 reflects parabolic oxidation kinetics: layer growth follows a square-root-of-time law, and the rate depends on local temperature alone through the Arrhenius term. Unlike nickel migration, interconnect oxidation is therefore not sensitive to current density. As $h_{\mathrm{IC-ox}}$ increases, the contribution $R_{\mathrm{IC-ox}}$ in Equation 3.21 rises and adds to the degradation summation in the total ohmic resistance. The constants k_{mg} and $E_{a,\mathrm{ox}}$ are fitted to experimental data [4], [5]. Together with nickel migration, this completes the ohmic degradation terms; the remaining mechanism acts on the cathode exchange current and is discussed next.

3.2.4 Nickel Agglomeration

A further degradation mechanism affecting the cathode exchange current is nickel agglomeration. Unlike the ohmic contributions discussed above, it is typically modelled not as a resistance but as a multiplier on λ_c, the normalized triple phase boundary [4]. Whether the underlying process is nickel migration or another phenomenon remains debated, and the mechanistic details are therefore not pursued here. Because agglomeration develops on a relatively short timescale, λ_c is assumed to have reached its asymptotic value of approximately 0.55 [4].

Description	**Equation**	**Ref.**
Total ohmic resistance	$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left\{\frac{E_{\text{act}}^{\text{ohm}}}{RT}\right\} + R_{\text{contact}} + \sum_{k\in\mathcal{D}_{\text{ohm}}} R_k$	
Nickel-migration resistance	$R_{\text{Ni-mi}} = \frac{d_{\text{Ni-mi}}}{(\frac{\varepsilon}{\tau}\sigma_{\text{elec}})}$	Equation 3.19
Electrode conductivity	$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,o}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right)$	Equation 3.20
Nickel thickness rate	$\frac{\partial d_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 iT + c_4$	[2]
Interconnect oxidation resistance	$R_{\text{IC-ox}} = \frac{d_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}}$	Equation 3.21
IC oxide conductivity	$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp\left\{-\frac{E_{\text{el}}}{RT}\right\}$	Equation 3.22
IC oxidation rate	$\frac{dd_{\text{IC-ox}}^2}{dt} = k_{\text{mg}} \exp\left\{-\frac{E_{a,\text{ox}}}{RT}\right\}$	Equation 3.23
Normalized triple phase boundary	$\lambda_c = 0.55$	[2]

Table 3.2: Overview of degradation mechanisms (ohmic contributions and slow-state dynamics). The base cell equations are summarized in Table 3.1.

3.3 Assumptions

The stack model in the preceding sections couples molar transport, electrochemistry, and an energy balance along a one-dimensional channel. The resulting relations are collected in Table 3.1; degradation is summarized in Table 3.2. The closed formulation rests on the simplifications listed below, which are introduced at various points in the derivation.

3.3.1 Geometry, flow, and thermodynamics

1. **Constant channel area.** The cross-sectional area A_{int} is uniform along the channel.
2. **One-dimensional flow.** Properties vary only in the axial direction z; width w and height d enter as geometric scalars.
3. **Ideal gas.** Gas mixtures obey the ideal-gas law.
4. **Negligible concentration overpotential.** Mass-transport limitations at the electrodes are not resolved; molar fractions on each electrode side are taken as uniform (Equation 3.8).
5. **Negligible heat conduction.** The conductive term $\nabla \cdot (\lambda_T \nabla T)$ is dropped in the energy balance (Equation 3.16).
6. **Constant pressure.** $\partial_t p \approx 0$ and $h \approx c_p T$ in the enthalpy formulation.
7. **Prescribed inlet flows.** Fuel and air volumetric flow rates are inputs, not optimized.

3.3.2 Electrochemistry and degradation

8. **Butler–Volmer kinetics.** Anodic and fuel-side (cathode) activation overpotentials follow Equation 3.12 with parameters from [13].
9. **Two dynamic ohmic degradation mechanisms.** Nickel migration and interconnect oxidation evolve via Table 3.2; other ageing paths are not modelled dynamically.
10. **Nickel agglomeration at steady state.** Because agglomeration develops on a short timescale relative to the optimization horizon, the cathode multiplier λ_c is fixed at ≈ 0.55 [4] rather than solved as an additional state.
11. **Spatially resolved degradation.** Slow states are stored per FVM cell ($\mathbf{d}_{\text{Ni-mi},i}, \mathbf{s}_{\text{IC-ox},i} \in \mathbb{R}^{N_{\text{cell}}}$); interconnect oxidation is advanced in squared-thickness form (Equation 4.18).

3.3.3 Quasi-steady coupling and time integration

12. **Quasi-steady cell physics.** At each time step the FVM steady-state problem (molar fractions and temperature along z) is solved for the current controls and degradation state before the slow ODEs are updated.
13. **Implicit Euler in time.** Degradation states are advanced with the implicit Euler scheme on a uniform grid (Equation 4.16).

4 Optimal Control and Solution Method

4.1 Optimal Control Problem

4.1.1 Introduction

With the cell model in place, and a nice degradation setup. Operation can be simulated for any admissible inlet conditions and cell voltage (within a reasonable range). What remains is to specify an economic objective and formulate the corresponding optimal control problem. Once again, related to the water gun problem, in this chapter, the objective of "hitting Thomas" is formulated.

Following discussions with Dynelectro and DTU Energy, and a review of how the objective is set in most literature [9],[3],[6], the general structure adopted is to maximize running revenue (hydrogen production while minimizing production cost) and to minimize degradation.

$$
\begin{aligned}
\max_{\mathbf{u}\in\mathcal{U}} \quad & \int_0^T R \,\mathrm{d}t + \Psi \\
\text{s.t.} \quad & \frac{\partial n_i}{\partial t} \text{ satisfies Table 3.1}, \qquad \text{for } i \in \mathcal{S} \\
& \frac{\partial d_i}{\partial t} \text{ satisfies Section 3.2.1}, \quad \text{for } i \in \mathcal{D}
\end{aligned}
\tag{4.1}
$$

Here $\mathbf{u}(t)$ collects the control inputs, that is the applied voltage $U(t)$ and the inlet temperature $T_{\text{in}}(t)$ that can be chosen as functions of time, and $\mathbf{d}$ collects the degradation state variables;

$$
\mathbf{d} = \Big[d_i(t)\Big]_{i\in\mathcal{D}} .
$$

Looking at Equation 4.1, the integrand $R(\mathbf{u}, \mathbf{d})$ is the **revenue rate**; maximizing its time integral corresponds to maximizing operating income over $[0, T]$. The term $\Psi(\mathbf{d})$ is the **salvation** contribution: a scalar that rewards the optimizer for how much useful life remains in the cell at the end of the horizon.

It is worth noting that minimizing degradation is in its core, a part of minimizing production cost: when a cell degrades the resistance rises, leading to higher power for the same hydrogen output, and in the worst case degradation leads to a malfunctioning cell, requiring replacement of the cell or, at facility scale, a stack. It should also be noted that using terminal degradation is a modeling choice; using the running degradation rate inside the integral would have been equally valid and may be simpler to implement.

4.1.2 Revenue

The revenue rate is taken in the explicit form shown in Equation 4.2.

$$
\begin{aligned}
R &= c_{\mathrm{H_2}} \Delta \dot{n}_{\mathrm{H_2}} - c_P P \\
\Delta \frac{\partial n_{\mathrm{H_2}}}{\partial t} &= A_{int} N_{out,\mathrm{H_2}} - \frac{\partial n_{in,\mathrm{H_2}}}{\partial t} \\
P &= P_{cell} + P_{heat}
\end{aligned}
\tag{4.2}
$$

Here $\frac{\partial n_{in,\mathrm{H_2}}}{\partial t}$ is specified as an input (part of $\mathbf{u}$), P is the total electrical power, P_{cell} the power supplied to the stack, and P_{heat} the auxiliary sensible heating power.

Thus, R is the value of net hydrogen production (outlet minus inlet, in appropriate units) minus the cost of power. All quantities in Equation 4.2 follow from the cell model summarized in Table 3.1.

From the model date the power of the cell is computed by $P_{cell} = I_{cell} U$ with:

$$
I_{cell} \approx \frac{A_{cell}}{N_{cell}} \sum_{k=1}^{N_{cell}} i_k .
$$

The heating power is the **sensible** auxiliary preheat duty: for each stream, the enthalpy difference between the **inlet** composition at the target inlet temperature T_{in} and the same composition at the recovered outlet temperature $T_{out} - T_{loss}$. Composition enthalpy of the electrochemical conversion is excluded from P_{heat} and is accounted for via P_{cell}. Positive P_{heat} means external preheat is required ($T_{in} > T_{out} - T_{loss}$); negative values are recovery credits.

$$
P_{\text{heat}} = \sum_{\zeta \in \{a,f\}} \dot{V}_\zeta \, A_{int} \left(\sum_{i \in \mathcal{S}_\zeta} \mu_{i,\text{in}} \, h_i(T_{in}) - \sum_{i \in \mathcal{S}_\zeta} \mu_{i,\text{in}} \, h_i(T_{out} - T_{loss}) \right)
\tag{4.3}
$$

Where outlet thermal energy is assumed recoverable toward the inlet stream with an absolute loss T_{loss}, and $\mu_{i,\text{in}}$ denotes inlet molar concentrations.

With that the running revenue or integrant is settled, hence we can look at the salvation.

4.1.3 Salvation

After operating up to time T, the running profit captures cash flow from hydrogen sales net of power cost. A salvation term is included, with the objective in mind, that the cell shouldn't be overused leading to, a cell which cannot gain its own cost in revenue. The idea is to measure, in a simple aggregate way, how close the cell is to end-of-life limits across degradation mechanisms that can cause failure.

Equal weight is assigned to each index in $\mathcal{D}$, leading to

$$
\Psi = \frac{c_{cell}}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(1 - \frac{d_i}{d_{i,\text{critical}}} \right)^{\alpha_\Psi}
\tag{4.4}
$$

Here c_{cell} scales the reward to the capital cost of the cell, and $\mathcal{D}$ is the set of degradation. Initially $\alpha_\Psi = 1$

For each i, the factor $\max\{0, 1 - d_i/d_{i,\text{critical}}\}$ is the unused fraction of the margin to the critical level at stop time T. Averaging over $\mathcal{D}_{\text{fail}}$ and scaling by c_{cell} yields a terminal value that increases when the cell stops farther from those limits.

4.1.3.1 Extra note on Salvation

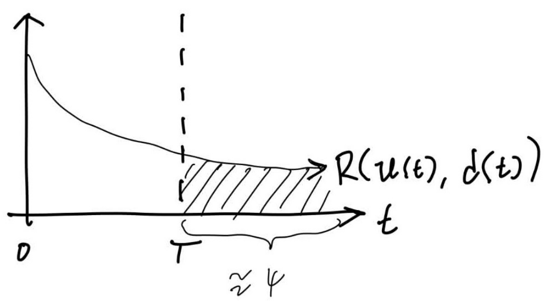


Figure 4.1: CAPTION

The ideal aim would be to extract the most value from a cell over its lifetime, which would correspond to optimizing the integral in Equation 4.1 with $\Psi = 0$ and T a conditioned variable equal to the lifetime. That horizon is not practical, since degradation and prices cannot be predicted that far ahead. Instead, the objective optimizes value over a short future horizon while retaining potential revenue. This is illustrated in Figure 4.1.

The objective is essentially the area under the revenue-rate curve. The formulation used here approximates maximization of that area over the full lifetime, but only up to the finite horizon T.

4.1.4 H_2 storage and delivery agreement

Beyond the simple revenue model in Equation 4.2, a contract-style extension adds buffer storage and a fixed hydrogen delivery obligation. The cell and degradation equations are unchanged; only the economic layer is extended. The contractual offtake is represented by a constant delivery rate $\dot{n}_{H_2,\text{agreement}}$. In practice, delivery may occur in batches; here it is approximated as a continuous flow over $[0, T]$. Shortfalls relative to production are met from inventory or, if the tank is empty, from spot purchases at rate $\dot{n}_{\text{purchase}}$, penalized in the revenue at $\kappa_{\text{buy}}\, c_{H_2}$. The resulting revenue rate is given in Equation 4.5.

$$R_{\text{tank}} = c_{H_2}\, \dot{n}_{H_2,\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{H_2}\, \dot{n}_{\text{purchase}}. \tag{4.5}$$

Inventory is modelled as a lumped storage tank. The net production surplus $\Delta\dot{n}$ is the difference between stack output and the agreement rate. Inventory increases only when production exceeds delivery; spot purchases activate only when production falls short. This gives Equation 4.6 and Equation 4.7.

$$\begin{aligned}
\Delta\dot{n} &= \dot{n}_{\text{prod}} - \dot{n}_{H_2,\text{agreement}} \\
\dot{m}_{\text{tank}} &= \max(0,\ \Delta\dot{n}) && (4.6)\\
\dot{n}_{\text{purchase}} &= \max(0,\ -\Delta\dot{n})\,. && (4.7)
\end{aligned}$$

At the end of the horizon the tank may still hold inventory. Two terminal treatments are used in the results.

In the first, remaining hydrogen is valued as liquidation on the open market, giving the additional term in Equation 4.8.

$$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\text{H}_2}\, m_{\text{tank}}(T), \tag{4.8}$$

where $\kappa_{\text{sell}} \approx \frac{1}{8}$ when end-of-horizon stock must compete with *black* hydrogen. In the second, a cyclic inventory constraint replaces terminal liquidation:

$$m_{\text{tank}}(0) = m_{\text{tank}}(T), \tag{4.9}$$

which is natural when the operating window is meant to repeat periodically.

Under Equation 4.9, the objective is invariant to a uniform upward shift of the inventory trajectory: if $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0$ is feasible, so is $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0 + a$ for any $a > 0$, provided the shifted trajectory remains admissible. To select the trajectory with the smallest inventory, a penalty term is subtracted from the objective:

$$-\min_t m_{\text{tank}}(t).$$

When the inventory stays strictly positive over $[0, T]$, this term equals the minimum inventory level and removes the shift degeneracy.

4.1.4.1 Full objective (tank mode)

When tank mode is active, the optimizer solves

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}}\,\mathrm{d}t + \Psi(\mathbf{d}(T)) + \Psi_{\text{tank}}(m_{\text{tank}}(T)) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = 0 \end{aligned} \tag{4.10}$$

or

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}}\,\mathrm{d}t + \Psi(\mathbf{d}(T)) - \min_t m_{\text{tank}}(t) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = m_{\text{tank}}(T) \end{aligned} \tag{4.11}$$

instead of Equation 4.1 with Equation 4.2. Fixed delivery revenue decouples cash flow from instantaneous production: the optimizer minimizes power cost and spot purchases while building inventory when production is favourable, subject to the usual salvation trade-off on degradation.

An overview of all relavant equations for the optimal control problem is stated in Table 4.1.

Description	Equation	Ref.
General objective function	$\int_0^T R, \mathrm{d}t + \Psi$	Equation 4.1
Revenue	$R = c_{\mathrm{H_2}} \Delta \dot{n}_{\mathrm{H_2}} - c_P P$	Equation 4.2
Salvation	$\Psi = \frac{c_{cell}}{\lvert\mathcal{D}\rvert} \sum_{i\in\mathcal{D}} \left(1 - \frac{d_i}{d_{i,\text{critical}}}\right)^{\alpha_\Psi}$	Equation 4.4
Tank balance	$\dot{m}_{\text{tank}} = \max\!\left(0,\ \dot{n}_{\text{prod}} - \dot{n}_{\mathrm{H_2},\text{agreement}}\right)$	Equation 4.6
Spot purchase	$\dot{n}_{\text{purchase}} = \max\!\left(0,\ \dot{n}_{\mathrm{H_2},\text{agreement}} - \dot{n}_{\text{prod}}\right)$	Equation 4.7
Contract revenue	$R_{\text{tank}} = c_{\mathrm{H_2}} \dot{n}_{\mathrm{H_2},\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{\mathrm{H_2}}\, \dot{n}_{\text{purchase}}$	Equation 4.5
Terminal liquidation (alternative A)	$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\mathrm{H_2}}\, m_{\text{tank}}(T)$	Equation 4.8
Cyclic inventory constraint (alternative B)	$m_{\text{tank}}(0) = m_{\text{tank}}(T)$	Equation 4.9
Minimum-inventory penalty (with alternative B)	$-\min_t m_{\text{tank}}(t)$	Equation 4.11
Cell Power	$P_{cell} = I_{cell} U$	
Heating Power	$P_{\text{heat}} = \sum_{\zeta\in\{a,f\}} \dot{n}_\zeta \left(\sum_{i\in\mathcal{S}_\zeta} x_i h_i\left(T_{in}\right) - \sum_{i\in\mathcal{S}_\zeta} x_i h_i\left(T_{out} - T_{loss}\right)\right)$	

Table 4.1: Overview of the optimal control formulation (objective, revenue, terminal value, and power bookkeeping). The underlying cell equations are summarized in Table 3.1; degradation is treated in Section 3.2.1. Optional tank economics (Section 4.1.4) add contract revenue and inventory dynamics; end-of-horizon inventory is closed either by terminal liquidation (alternative A) or by a cyclic constraint with a minimum-inventory penalty (alternative B).

4.1.5 Economic assumptions

The optimal-control formulation layers an economic objective on top of the stack model. Its governing expressions—objective, revenue rate, salvation, and optional tank bookkeeping—are collected in Table 4.1. The following assumptions apply to that layer and to its piecewise-constant time discretization.

1. **Piecewise-constant controls.** U and T_{in} are constant on each of the N_t control intervals (Section 4.2.6).
2. **Recoverable outlet enthalpy.** Auxiliary heating power P_{heat} credits recovery of outlet sensible heat toward the inlet stream, minus a fixed loss T_{loss} (Equation 4.2).
3. **Finite rolling horizon.** The objective maximizes revenue and salvation over a chosen horizon T, not the full stochastic stack lifetime (Figure 4.1).
4. **Tank scenarios (when active).** Buffer storage is unlimited; contractual delivery $\dot{n}_{\text{H}_2,\text{agreement}}$ is constant; shortfalls are covered by inventory or spot purchase at $\kappa_{\text{buy}} c_{\text{H}_2}$ (Section 4.1.4).

4.2 Building the numerical solver

4.2.1 Introduction

This chapter presents the construction of numerical tools required to determine the optimal control $\mathbf{u}^*(t)$ for the system. Unlike simple textbook examples, where a Wikipedia lookup was sufficient, the problem considered here is more complex and necessitates a structured, step-by-step methodology. The central question addressed is: What tools and methods are needed to solve the optimal control problem outlined in Section 4.1.1?

The overall strategy is as follows:

1. **Steady-state cell equations:**
   As the focus is on long-term behavior, it is assumed that the dynamics described by the differential equations in Table 3.1 occur much faster than the overall optimization timescale, treating them as instant. This allows for replacement with a set of steady-state equations, which are simpler to solve. The Finite Volume Method (FVM) is used to solve these equations numerically.

2. **Degradation dynamics:**
   With the steady-state cell solver established, its outputs serve as algebraic inputs to the degradation model (Table 3.2). This model captures the system's degradation over time through differential equations, which are solved numerically using the implicit Euler method.

3. **Objective and optimal control:**
   The optimal control objective (Table 4.1) is addressed by discretizing the control input with the single shooting method. In this approach, both the revenue and degradation equations are integrated forward in time as a system of differential equations.

Structuring the problem in these three parts (steady-state solution, degradation modeling, and control optimization) enables a systematic approach to the full optimal control challenge.

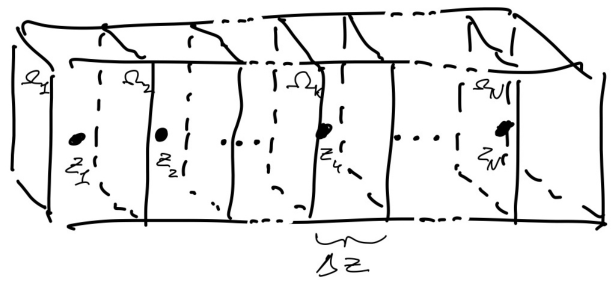


Figure 4.2: A sketch of the finite volume method. A large volume, corresponding to the air and gas channel, is split into smaller volumes. The greater volume is divided into N volumes, each represented by the midpoint, yielding a 1D system instead of 3D. Integration over the densities in each small volume gives the mass in that volume. The figure also introduces z_k (midpoint), Ω_k (small volume), and Δz (length of the volume).

4.2.2 Finite volume method (FVM)

With the descriptive equations in place (Table 3.1, Section 3.2.1, and Section 4.1.1), a numerical solution method is required. All equations can be written in the general form

$$\frac{\partial u}{\partial t} = Q_u - \frac{\partial F_u}{\partial z} \tag{4.12}$$

where Q_u denotes the source term and F_u the flow term; both can be computed directly from the state. For now only the steady-state solution is of interest, which simplifies the problem considerably and yields the following *boundary value problem*:

$$0 = Q_u + \frac{\partial F_u}{\partial z}$$

The *finite volume method* is used to solve this steady-state BVP [8].

As shown in Figure 4.2, the domain is split into smaller subdomains coupled by flux from one volume into the next. Integration of the density u over a small volume Ω_k of volume V_k proceeds as follows.

$$\int_{\Omega_k} \frac{\partial u}{\partial t} = \int_{\Omega_k} Q_u \, dV - \int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV$$

The source term is approximated by a rectangle (midpoint rule): $\int_{\Omega_k} Q_u \approx Q_u(z_k) A_{int} \Delta z$. For the flux term, the fundamental theorem of calculus is used to evaluate the antiderivative in the z-direction, with the assumption that the concentration is constant perpendicular to z, giving:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV = A_{int} F_u \Big|_{\partial \Omega_k} = A_{int} \Big(F_u(z_{k+\frac{1}{2}}) - F_u(z_{k-\frac{1}{2}}) \Big)$$

The boundary values $F_u(z_{k-\frac{1}{2}})$ are approximated by the value in the previous cell, consistent with the flow direction of the system (in cell k there is no information from cell $k+1$, but there is from cell $k-1$). The integral is therefore approximated as:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \approx \Big(F_u(z_k) - F_u(z_{k-1})\Big) A_{int}$$

Combining the source and flux approximations gives the discrete equation Equation 4.13.

$$\frac{\partial}{\partial t} \int_{\Omega_k} u \, dV = Q_u(z_k) A_{int} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1})\Big) A_{int} \tag{4.13}$$

or at steady state:

$$0 = Q_u(z_k) A_{int} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1})\Big) A_{int}$$

4.2.2.1 Applying boundary conditions

The problem has two boundary conditions. The first is a Dirichlet condition at the inlet, imposed by defining the first flux as:

$$F_u(z_{-1}) = F_u(z_{inlet})$$

For the Neumann condition at the outlet, no change is made to the equations, since the upstream flux in the FVM makes it possible to neglect a Neumann boundary term at the end.

4.2.2.2 FVM on specific equations

Applying the FVM to the governing equations yields the following discretization; see Appendix A3 for derivations.

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) \approx w N_{c,i} \Delta z - A_{int}(N_i(z_k) - N_i(z_{k-1})), \tag{4.14}$$

And

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T A_{int} \Delta z - A_{int} \sum_{i \in \mathcal{S}} v_i c_{p,i} \Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big) \tag{4.15}$$

4.2.3 FVM model validation and Grid independence

4.2.3.1 IV-curve validation

After implementing all the equations given in Table 3.1, the model was validated against an IV-curve from experimental data [16]. The error is below 10^{-2} except at quite high voltage/current, where the error is larger. Figure 4.3 compares the model with the data.

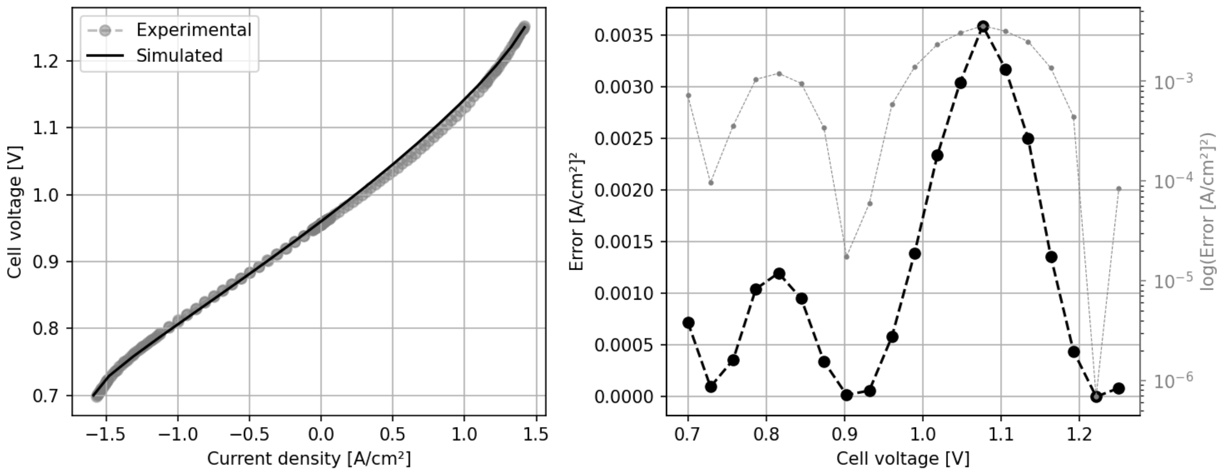


Figure 4.3: Model validation by comparing to an IV-curve from experimental data. The simulated current–voltage curve is compared to the experimental curve on the left. On the right, the error in current is shown as a function of voltage, on both logarithmic and linear scales. The experimental data is from [14], Fig. 2.

This level of agreement is sufficient for the intended operating range, as very high voltages are not relevant for typical operation. The main variable of interest is the change in moles across the cell, which exhibits similar trends; at high voltage or current, the error in molar change decreases further, since the molar fractions approach their extrema (0 or 1), making relative errors negligible.

4.2.3.2 Grid independence

Grid refinement was used to assess how much the states change when the mesh is refined, and thereby to estimate the resolution required for reliable simulations later. The steady-state solver was run while doubling the number of grid points each time. Figure 4.4 indicates that if a relative error target of approximately 10^{-3} was the target. The amount of volumes needed would be around 200. But for this thesis, the amount of volumes will be 50, meaning that a relative error of around 10^{-2} will have to suffice.

But nonetheless the Steady-state equations can be solved to a rather fine precision. Leading to the next step: Solving the ODE's using the outputs of the output of the FVM.

4.2.4 Implicit Euler method (ODE solver)

As stated in the introduction to this chapter, the FVM gives the algebraic outputs needed to solve the differential equations that define degradation (Table 3.2). Those ODEs are advanced with the implicit Euler method on a uniform time grid [10].

4.2.4.1 General form

Take the ODE: $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$. Discretize time on $t \in [0, T]$ into N_t steps with $\mathbf{x}_i = \mathbf{x}(t_i)$, $t_i = t_{i-1} + \Delta t_i$, $t_0 = 0$, and $t_{N_t} = T$. Given an initial state $\mathbf{x}_0$, the implicit Euler update is

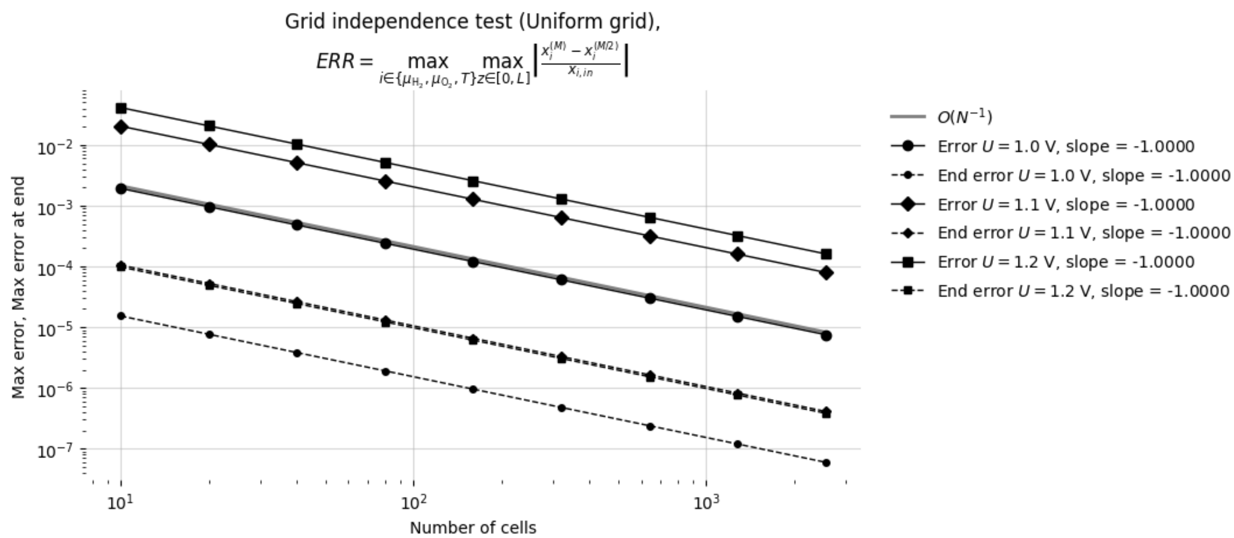


Figure 4.4: Grid-independence study of the steady-state FVM solver. The number of cells N is doubled repeatedly $(5, 10, ...)$ at fixed inlet conditions and three cell voltages ($U = 1.0$, 1.1, and 1.2 V). Solid markers show the maximum relative change in hydrogen and oxygen chemical potentials and temperature between successive refinements; dashed markers show the same quantity at the channel outlet. All curves fall approximately as $O(N^{-1})$ (grey reference). At $N \approx 200$ the error reaches $\sim 10^{-3}$; the thesis adopts $N = 50$, for which the error is $\sim 10^{-2}$.

$$\mathbf{x}_i = \mathbf{x}_{i-1} + \Delta t_i \, \mathbf{f}(t_i, \mathbf{x}_i), \qquad i = 1, \ldots, N_t. \tag{4.16}$$

Because $\mathbf{x}_i$ appears on both sides, Equation 4.16 is a fixed-point equation at each step. Damped fixed-point iteration is therefore a natural solver. The update is also Markovian: step i depends only on $\mathbf{x}_{i-1}$ and the control on $[t_{i-1}, t_i)$, so the integration can proceed one time step at a time.

4.2.4.2 Specific ODE setup

For this thesis the vector field $\mathbf{f}$ is expensive: evaluating $\dot{\mathbf{d}}$ requires converged FVM fields (temperature and current density along the stack), which themselves depend on the degradation state. The degradation state advanced at each control interval is

$$\mathbf{d}_i = \begin{bmatrix} \mathbf{d}_{\mathrm{Ni-mi,i}} \\ \mathbf{s}_{\mathrm{IC-ox,i}} \end{bmatrix}, \tag{4.17}$$

where $\mathbf{d}_{\mathrm{Ni-mi,i}}, \mathbf{s}_{\mathrm{IC-ox,i}} \in \mathbb{R}^{N_{\mathrm{cell}}}$ collect the per-cell slow states at time t_i, with $\mathbf{s}_{\mathrm{IC-ox,i}} = \mathbf{d}^2_{\mathrm{IC-ox,i}}$. Nickel migration stores layer thickness $d_{\mathrm{Ni-mi}}$ directly; interconnect oxidation stores squared oxide thickness $s_{\mathrm{IC-ox}} = d^2_{\mathrm{IC-ox}}$, consistent with Equation 3.23 and $R_{\mathrm{IC-ox}} = d_{\mathrm{IC-ox}}/\sigma_{\mathrm{IC-ox}}$ with $d_{\mathrm{IC-ox}} = \sqrt{s_{\mathrm{IC-ox}}}$.

With piecewise-constant controls $\mathbf{u}_i$ on $[t_{i-1}, t_i)$, implicit Euler gives

$$\begin{aligned} \mathbf{d}_{\text{Ni-mi,i}} &= \mathbf{d}_{\text{Ni-mi,i-1}} + \Delta t_i\, \dot{\mathbf{d}}_{\text{Ni-mi}}(\mathbf{d}_i, \mathbf{u}_i), \\ \mathbf{s}_{\text{IC-ox,i}} &= \mathbf{s}_{\text{IC-ox,i-1}} + \Delta t_i\, \dot{\mathbf{s}}_{\text{IC-ox}}(\mathbf{d}_i, \mathbf{u}_i), \end{aligned} \qquad i = 1, \dots, N_t, \tag{4.18}$$

where $\mathbf{d}_i$ stacks all per-cell degradation components at time t_i. The rates follow Table 3.2: $\dot{d}_{\text{Ni-mi,k}} = c_1 T_k + c_2 i_k + c_3 T_k i_k + c_4$ depends on local temperature T_k and current density i_k, while $\dot{s}_{\text{IC-ox,k}} = k_{\text{mg}} \exp\{-E_{a,\text{ox}}/(RT_k)\}$ depends on T_k only.

Those FVM outputs are not explicit inputs to the ODE; they are obtained by running the steady-state FVM solver with control $\mathbf{u}_i$ and degradation state $\mathbf{d}_i$. Because $\mathbf{d}_i$ appears inside that solve, each time step is a coupled fixed-point problem: guess $\mathbf{d}_i$, run FVM, evaluate the rates in Equation 4.18, update $\mathbf{d}_i$, and iterate until convergence. Giving us the heavy computation of solving the steady state for each fixed-point iteration, leading to a large computational time.

In general, the problem can be viewed on a two-dimensional grid: each spatial row (FVM cell) must be resolved repeatedly within a time step before advancing to the next column (time index).

4.2.5 Validation of the ODE solver

With the steady-state FVM validated in the previous section, the next step is to check that the coupled degradation integrator in Equation 4.18 reproduces expected long-term behaviour.

The reference case follows the potentiostatic traces reported by Beyrami et al. [4] (Fig. 8): fixed cell voltage $U = 1.285\,\text{V}$, inlet temperature $T_{\text{in}} = 750\,^\circ\text{C}$, and fuel composition $10\,\%$ H_2 / $90\,\%$ H_2O. At each time knot the FVM steady-state problem is solved, the degradation rates from Table 3.2 are evaluated, and the states are updated with implicit Euler (Equation 4.16) until $t = 25\,\text{kh}$. This produces a history of how the degradation mechanisms develop over time. From these quantities, the area-specific resistances (ASR) can be computed and compared with the traces from Beyrami et al. in Figure 4.5.

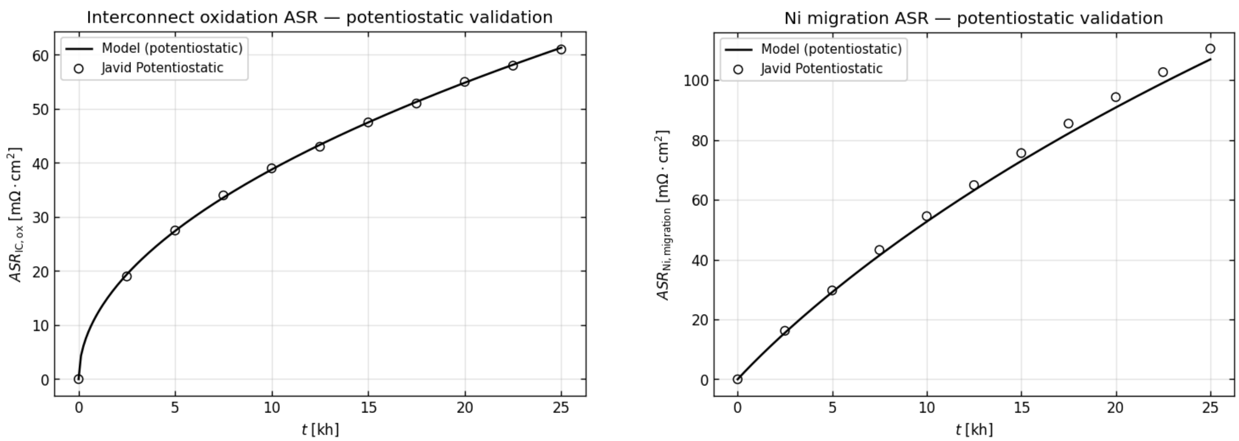
The interconnect-oxidation agreement is very good over the full 25 kh horizon. Nickel migration follows the same trend but sits slightly below the reference curve. Given the digitization uncertainty and the simplified inlet specification relative to the original study, this level of agreement is considered sufficient for the degradation model to be used in the optimal-control setup in Section 4.2.6.

4.2.6 Solving the Optimal Control Problem

The optimal control problem (Equation 4.1) is solved using a nonlinear programming (NLP) solver. The problem is set up for single shooting so that an NLP solver can find the optimum. The control function $\mathbf{u}$ is taken to be piecewise constant (as would also be the case in real applications).

$$\mathbf{u}^{(k)}(t) = \left\{ \mathbf{u}_i^{(k)} \quad t \in [t_i, t_{i+1}) \right.$$

The problem is split into N_t subproblems, each with constant control input. From Equation 4.1 the objective φ can be rewritten as:



(a) Interconnect oxidation. At $t = 25\,\mathrm{kh}$: model $61.3\,\mathrm{m\Omega \cdot cm^2}$, reference $61.0\,\mathrm{m\Omega \cdot cm^2}$. RMSE $= 0.26\,\mathrm{m\Omega \cdot cm^2}$.

(b) Nickel migration. At $t = 25\,\mathrm{kh}$: model $106.9\,\mathrm{m\Omega{\cdot}cm^2}$, reference $110.5\,\mathrm{m\Omega \cdot cm^2}$. RMSE $= 2.45\,\mathrm{m\Omega \cdot cm^2}$.

Figure 4.5: Potentiostatic validation of ASR for (a) interconnect oxidation and (b) nickel migration. Black line: implicit-Euler simulation. Open markers: digitized reference data from [2].

$$
\begin{aligned}
\varphi &= \int_0^T R(\mathbf{u}, \mathbf{d})\ \mathrm{d}t + \Psi(\mathbf{d}) \\
\Rightarrow \dot{\varphi} &= R(\mathbf{u}, \mathbf{d})
\end{aligned}
\tag{4.19}
$$

Thus φ is obtained by solving the auxiliary differential equation with initial condition $\varphi(0) = 0$ (numerical integration), concurrently with the state equations, and adding $\Psi(\mathbf{d})$ at the final time when the degradation at $t = T$ is known.

For the constraints in Equation 4.1, which are equality constraints, they are used directly to compute degradation and molar flows, leaving only $\mathbf{u}$ as a decision variable for the optimizer. The NLP solver also requires the gradient $\nabla_{\mathbf{u}}\varphi$; adjoint gradient methods are used [15], [7]. Further details are in Appendix A5; here it suffices that $\nabla_{\mathbf{u}}\varphi$ is available.

4.2.6.1 The full setup

The discretized problem solved is:

$$
\begin{aligned}
&\max_{\mathbf{u}} && \varphi_N + \Psi(\mathbf{d}_N) \\
&\text{s.t.} && \varphi_{i+1} = \varphi_i + \dot{\varphi}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \ldots N-1 \\
& && \mathbf{d}_{i+1} = \mathbf{d}_i + \dot{\mathbf{d}}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \ldots N-1 \\
& && \varphi_1 = 0, \\
& && \mathbf{d}_1 = \mathbf{o}.
\end{aligned}
\tag{4.20}
$$

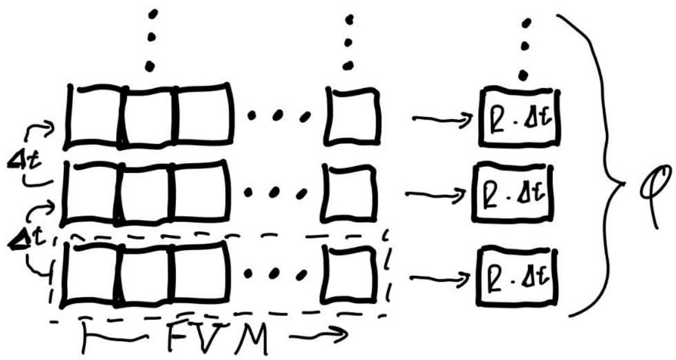


Figure 4.6: Overview of the single-shooting discretization. The horizon is divided into intervals of length Δt (vertical axis). At each interval a piecewise-constant control $\mathbf{u}_i$ drives a steady-state FVM solve of the cell equations (spatial cells along the stack, dashed box). The resulting fields and degradation state feed the running objective rate $R(\mathbf{u}, \mathbf{d})$, whose contribution $R\,\Delta t$ is accumulated forward to give the total objective φ.

as a single shooting problem, with **d** computed from the constraint equations. A convex programming algorithm is used to find the optimal u; the problem is generally convex. For this project the trust-constr method from SciPy is used [19], a trust-region method that employs the computed gradient of φ.

5 Results and Findings

5.1 Introduction

With the full machinery in place, it was time to put it to work. The preceding chapters set up the model, the objective, and the solver; what was still missing before any results could be discussed was a single list of the **numerical settings** actually used in the optimization runs. Those defaults are collected here. Later sections vary individual knobs; this section is the common baseline they all share unless stated otherwise.

The objective itself is unchanged from Equation 4.1: maximize integrated revenue plus terminal salvation over a finite horizon. The prices and break levels are the concrete numbers plugged into that formulation. Unless stated otherwise, the runs use the settings in Table 5.1. Literature values for stack geometry and flows [16], end-of-life limits [2], cell cost [3], and energy prices [1] are indicated in the reference column; entries marked *Chosen* are project defaults.

Several entries in Table 5.1 are marked as chosen rather than taken from the literature. Brief rationale is given here; fuller discussion of all *Chosen* settings across the results chapter is in Section 6.4. The voltage and inlet-temperature bounds keep the optimizer within a range of physically plausible operating conditions for the stack model. The number of FVM cells N_{cell} and the control segments N_t were set by the computational budget of the project: finer discretizations were possible in principle, but the selected values keep each forward solve and outer iteration within tractable run times on the available hardware.

The 200-day horizon is chosen for the same practical reason. It is long enough that a change in the optimal control profile can show up within the optimization window, without extending T so far that run times and degradation forecasts dominate the workflow.

The parameter α_Ψ in Equation 4.4 controls the sharpness of the terminal penalty as the cell approaches its end-of-life degradation threshold. A value of $\alpha_\Psi = 0.5$ is chosen here, making the penalty function relatively gentle when the stack is healthy, but causing it to increase steeply as degradation nears its critical limit. This ensures the optimizer is sensitive to imminent failure while still allowing for some degradation earlier in the lifetime. Now to illustrate this the salvation for only a single degradation parameter is plotted in Figure 5.1, for different values of $\alpha_{\text{salvation}}$. Here we see that in $\alpha_{\text{salvation}} = 1$, we simply have a linear function. And as $\alpha_{\text{salvation}} \to 0$ we get a function that tends to zero loss, besides in the point where it hits the breaking point, where it has full loss. Here it was assessed that $\alpha_{\text{salvation}} = 0.5$ is a good compromise, between weighing all degradation while having that the degradation close to failure is of most importance.

Now through this we run 3 scenarios and analyze every scenario individually afterward generating further 3 results from sensitivity analisys.

Setting	Baseline value	Unit	Reference
Horizon T	200	days	Chosen
Control segments N_t	5	—	Chosen
U bounds	$[0.8,\ 1.5]$	V	Chosen
T_{in} bounds	$[700,\ 900]$	°C	Chosen
FVM cells N_{cell}	20	—	Chosen
Fuel / air flows	24 / 50	NL/h	[14]
Initial $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	0.75 / 1.5	μm	Values after 100 days burn-in
Break $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	5 / 10	μm	[18]
c_{H_2}	7.44	EUR/kg	[19]
c_P	0.079	EUR/kWh	[19]
c_{cell}	1000	EUR/m^2	[14], [1]
α_Ψ	0.5	—	Chosen
Optimization solver	SLSQP/Trust-Constr	—	—

Table 5.1: Baseline numerical settings for the optimization results in this chapter (unless a subsection states otherwise).

Scen.	Description	Key parameter varied	Avg. revenue [c/d]	Avg. salv. loss [c/d]	Avg. net econ. [c/d]
1	Baseline run	Current I	8.65	0.45	8.20
2	Constant delivery rate and tank storage	κ_{sell}, tank schedule	5.82	0.21	5.72
3	24 h cyclic power price	T, I	7.40	0.190	7.20

Table 5.2: Overview of the three result scenarios analysed in this chapter. Integrated revenue, salvation loss ($\Psi_0 - \Psi$), and net economy ($J - \Psi_0$) from Section 5.2, Section 5.3, and Section 5.4 are divided by the scenario horizon (200 days for scenarios 1–2, 24 h for scenario 3). Daily averages are reported in euro cents per day (c/d).

Optimization trajectories share a common multi-panel layout (fine forward replay): **Outside factors** (tank and, where applicable, c_P and $c_P P$), then **H_2 production**, **Degradation**, **Control dependencies**, and **Controllable variables** (the baseline omits **Outside factors**). Result figures use float page placement at full text width.

5.2 Baseline optimization result

As announced in Section 5.1, the first result is a baseline in which both controls are held constant over the full horizon: a single cell voltage U and a single inlet temperature T_{in}, broadcast to all $N_t = 5$ segments. The optimizer is run in potentiostatic-isothermal (pot-iso) mode, i.e. with fixed voltage and inlet temperature on every segment. This is the most restricted operating mode considered (U and T_{in} each collapse to one scalar decision variable), analogous to fixing both aim and pump pressure in the water-gun example from Section 2.1.

All other settings follow Table 5.1. The NLP returns

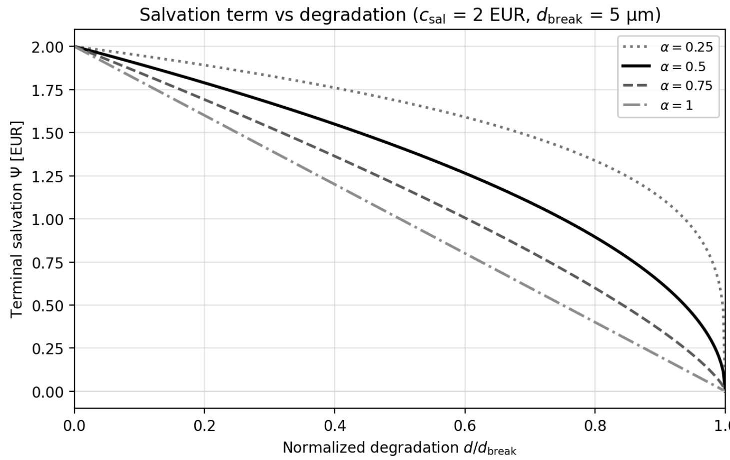


Figure 5.1: Effect of the salvation penalty sharpness parameter α_Ψ in Equation 4.4 for a single degradation variable. For $\alpha_\Psi = 1$ (blue, linear) all degradation is penalized equally, while lower α_Ψ (red, green) create a steeper penalty as end of life is approached, with near-zero penalty until close to the critical threshold. The value $\alpha_\Psi = 0.5$ (green) is chosen as a compromise.

$$U^* \approx 1.32\,\mathrm{V}, \qquad T_{\mathrm{in}}^* \approx 751\,^\circ\mathrm{C}\ (1024\,\mathrm{K}).$$

Figure 5.2 shows the replayed trajectory at these controls. Mean nickel-migration thickness grows from $0.75\,\mu$m to the $5\,\mu$m break limit; interconnect oxidation from $1.5\,\mu$m to about $3.6\,\mu$m. Running revenue accumulates over the 200-day window while nickel migration reaches the enforced break limit at $t = T$.

The accumulated revenue is 17.30 EUR and the terminal salvation term is $\Psi \approx 1.01$ EUR. Because the degradation states are seeded from a 100-day burn-in (Table 5.1), the salvation value at the horizon start is already $\Psi_0 \approx 1.91$ EUR rather than the pristine-cell value 2.00 EUR; the corresponding salvage loss before the optimization period is ≈ 0.09 EUR. Over the 200-day window the cell loses a further $\Psi_0 - \Psi \approx 0.90$ EUR of salvage value, giving a net economy $J - \Psi_0 \approx 16.40$ EUR on the $20\,\mathrm{cm}^2$ cell.

The optimum sits in exothermal rather than thermoneutral operation: the chosen voltage exceeds the thermoneutral value at the inlet temperature, so reaction and ohmic heat raise the local temperature from inlet to outlet (roughly 751 to $780\,^\circ$C at $t = T$). Figure 5.3 resolves this behaviour along the channel at $t = 0$, $t = T/2$, and $t = T$. Nickel migration is the life-limiting mechanism—by $t = T$ the thickness is within a few percent of the $5\,\mu$m cap everywhere along z, whereas interconnect oxidation remains well below its $10\,\mu$m limit and retains a visible outlet bias. The near-uniform nickel profile at end of life is a direct consequence of the exothermal temperature field: degradation rates depend on local temperature, and the monotonic heating along the channel homogenizes nickel growth even though the spatial initial condition at $t = 0$ is not flat. Interconnect oxidation, by contrast, still concentrates toward the hotter outlet.

The next thing to do was to finally give the control more freedom and let the individual control segments take whatever value optimized the most. Here a somewhat boring thing happened. The control did not change and the potentiostatic-isothermal operation is the best. Even though it is a boring result, it makes perfect sense: the unrestricted baseline still favors steady pot-iso controls, with voltage and inlet temperature locked

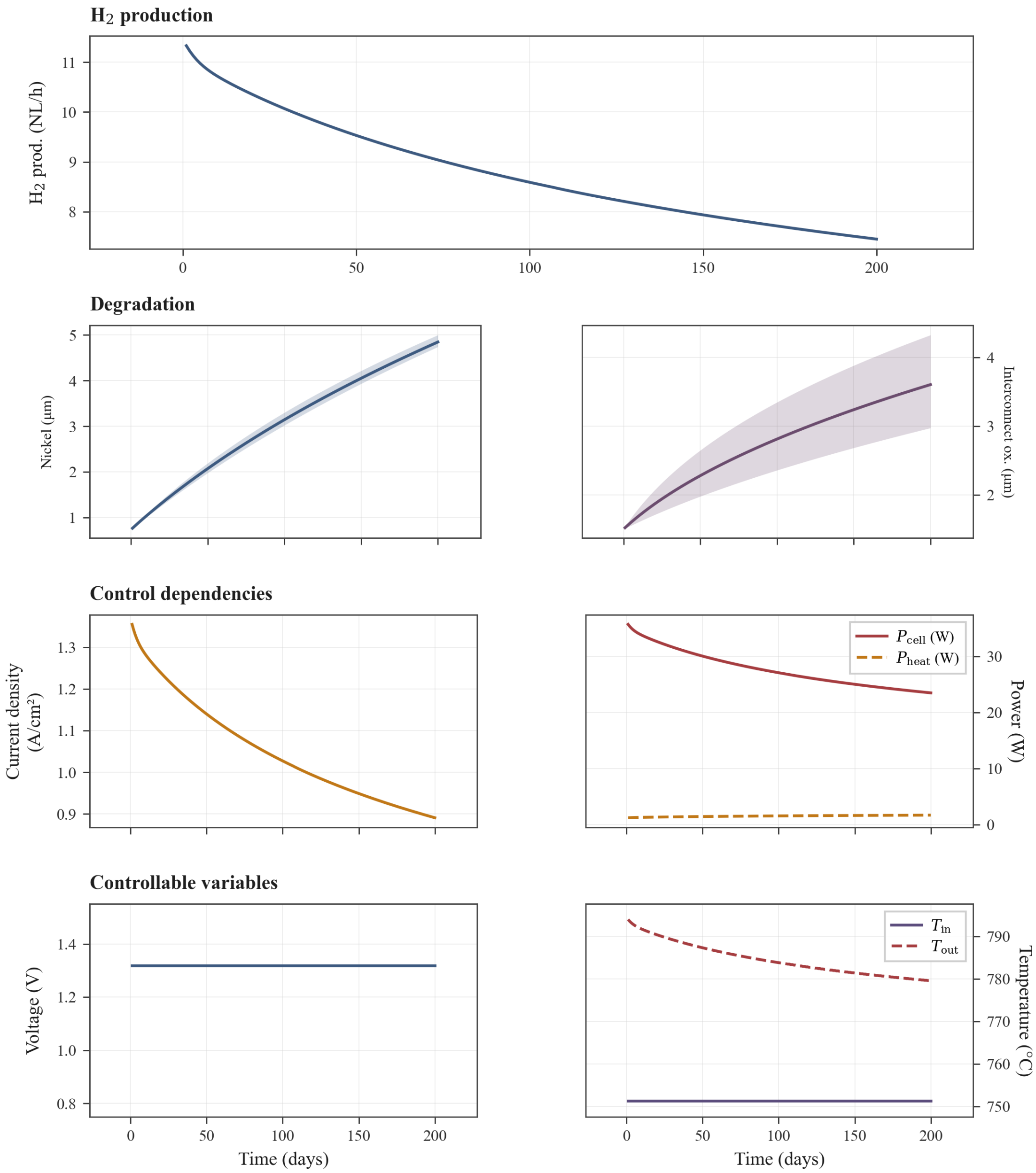


Figure 5.2: Baseline potentiostatic-isothermal optimum with no delivery contract ($N_t = 5$; Table 5.1). Single controls $U^* \approx 1.32\,\mathrm{V}$ and $T^*_{\mathrm{in}} \approx 751\,^\circ\mathrm{C}$ are held over 200 days. Exothermal operation (U^* above the thermoneutral voltage at T^*_{in}) sustains positive P_{heat} and a rising temperature profile along the cell; $\dot{n}_{\mathrm{H_2,prod}}$ falls from ≈ 11.3 to $7.5\,\mathrm{NL/h}$ as nickel thickness approaches the break cap. Net economy $16.40\,\mathrm{EUR}$ on the $20\,\mathrm{cm}^2$ cell.

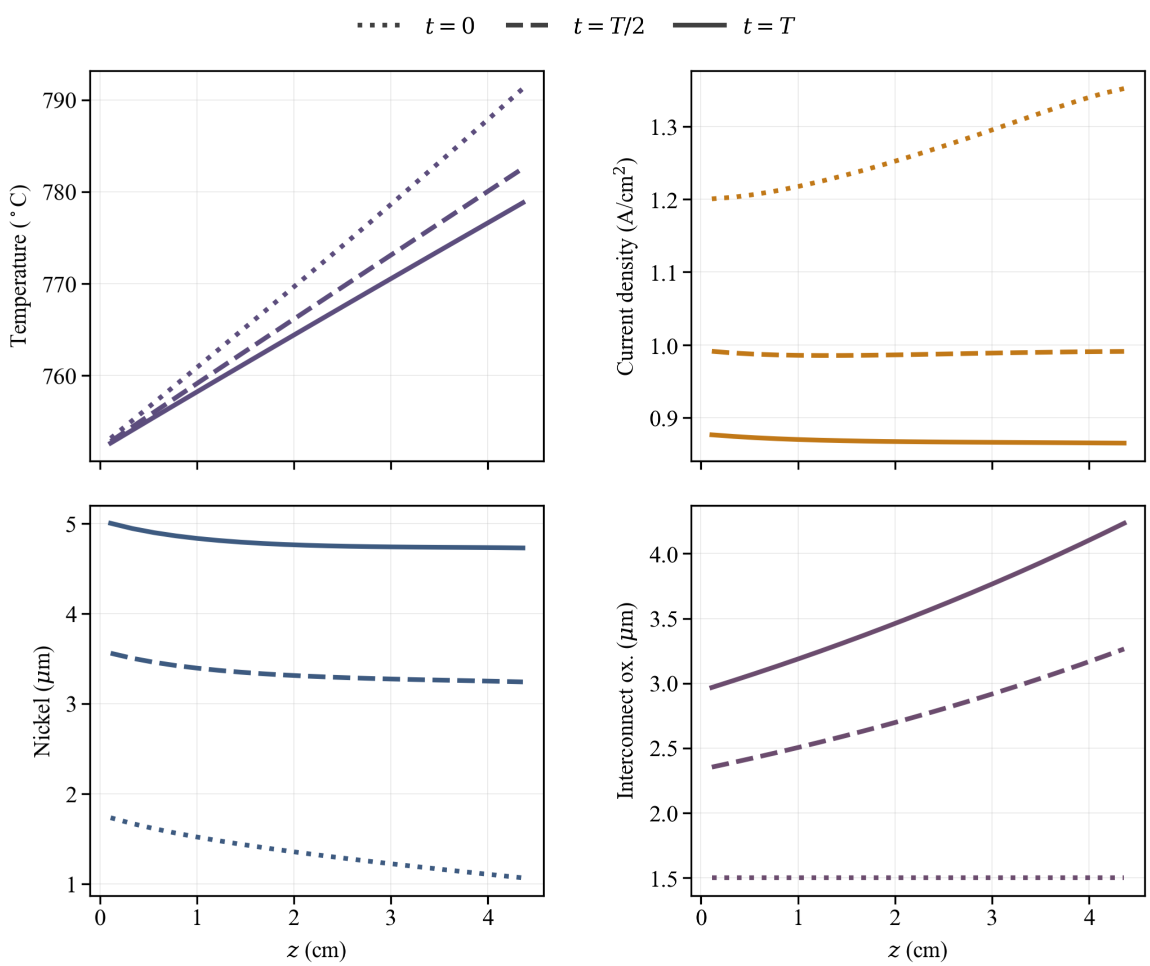


Figure 5.3: Spatial channel profiles at $t = 0$, $t = T/2$, and $t = T$ for the baseline pot-iso optimum (fine forward replay). Temperature rises along z under exothermal operation; nickel migration approaches the $5\,\mu$m break limit almost uniformly at $t = T$, while interconnect oxidation stays below its cap with stronger outlet weighting.

at the exothermal optimum that extracts revenue up to the nickel-migration break limit. So to make use of the dynamic flexibility of the model

	J	ϕ	ψ	ψ_{diff}	Net
No Cap	13.923	12.361	1.569	0.341	12.02
Cap	13.878	12.305	1.566	0.344	11.96

Table 5.3: Summary of economic and terminal values for constant-delivery (with and without tank cap). J: total value; ϕ: running revenue; ψ: terminal salvation; ψ_{diff}: difference in salvation; Net: net economy.

5.3 Constant delivery rate

The next result uses the storage-tank implementation from Section 4.1.4 with a fixed contractual delivery rate. All other settings follow Table 5.1; the additional tank parameters are listed in Table 5.4. The optimizer is run in **potentiostatic–isothermal** mode (U and T_{in} each constant over the horizon, broadcast to all $N_t = 20$ segments), matching the κ_{sell} sensitivity infrastructure in Section 6.4.1.

The agreement rate $\dot{n}_{\mathrm{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$ per cell sits below the unconstrained baseline production at $t = 0$ (Section 5.2), so surplus hydrogen can be stored rather than sold immediately. Scaled to a production setup of two stacks with about 150 cells each, the combined daily delivery is approximately $4.3\,\text{kg}\,\mathrm{H}_2$. The spot-purchase penalty $\kappa_{\text{buy}} = 1.5$ fines under-delivery at 50% of the contract hydrogen price. The terminal liquidation credit $\kappa_{\text{sell}} = \frac{1}{9}$ values leftover inventory at roughly 11% of the contract price, reflecting competition with lower-cost *black* hydrogen on an open market.

5.3.1 Pot-iso optimum at low liquidation credit

With $\kappa_{\text{sell}} = \frac{1}{9}$, terminal resale is so weak that the optimizer prefers to **hoard** surplus hydrogen rather than produce aggressively and liquidate at the horizon. Figure 5.4 shows the SLSQP optimum at this liquidation credit: a single pot-iso pair

$$U^* \approx 1.289\,\text{V}, \qquad T_{\text{in}}^* \approx 750\,^\circ\text{C},$$

held for all 200 days. Initial production ($\approx 8.1\,\text{NL/h}$) exceeds the agreement rate, so the tank fills monotonically to a peak of $\approx 0.27\,\text{kg}$ around day~170. As degradation lowers production below $6.5\,\text{NL/h}$, the inventory buffer covers the shortfall; no spot purchases occur. Because unlimited storage is free in the model and κ_{sell} is small, the optimizer does not schedule galvanostatic intervals—constant gentle operation minimizes salvation loss while accumulating inventory that is effectively worthless to sell.

The accumulated revenue is $\approx 11.64\,\text{EUR}$ and the terminal salvation term is $\approx 1.50\,\text{EUR}$; the terminal liquidation credit contributes only marginally. The net economy is $\approx 13.36\,\text{EUR}$ on the $20\,\text{cm}^2$ cell over 200 days.

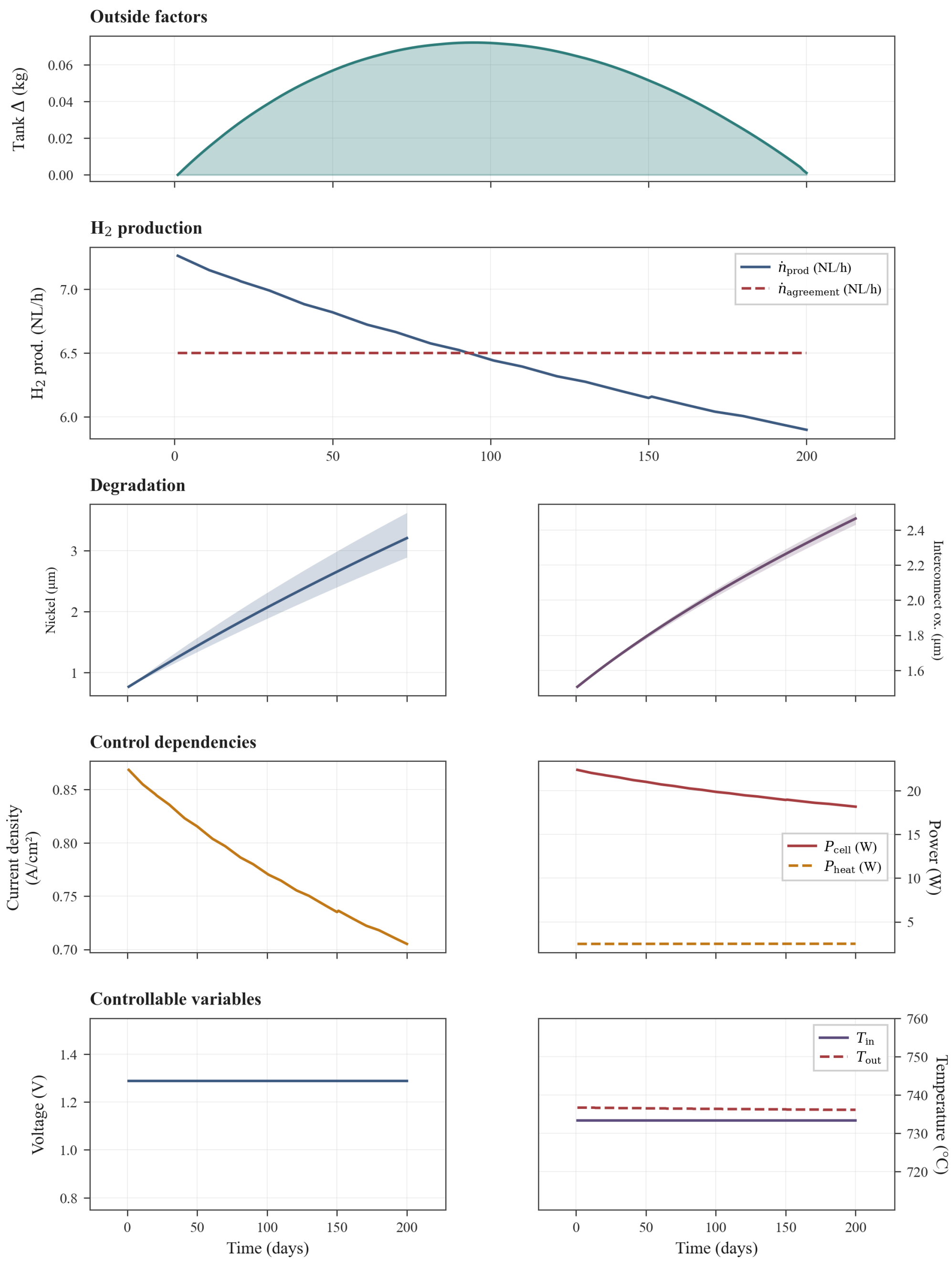


Figure 5.4: Pot-iso constant-delivery optimum at $\kappa_{\text{sell}} = \frac{1}{9}$ and $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$ ($N_t = 20$; Table 5.4). Single controls $U^* \approx 1.289\,\text{V}$, $T_{\text{in}}^* \approx 750\,^\circ\text{C}$; tank inventory peaks at $\approx 0.27\,\text{kg}$ (day 170) and remains near $\approx 0.26\,\text{kg}$ at the horizon. Running revenue $\approx 11.64\,\text{EUR}$, terminal salvation $\approx 1.50\,\text{EUR}$, net economy $\approx 13.36\,\text{EUR}$.

Setting	**Value**	**Unit**	**Reference**
Contractual delivery $\dot{n}_{\mathrm{H_2},\mathrm{agreement}}$	6.5	NL/h	Chosen
Initial tank inventory $m_{\mathrm{tank},0}$	0	kg	Chosen
Spot purchase penalty κ_{buy}	1.5	—	Chosen
Terminal liquidation credit κ_{sell}	$\frac{1}{9}$	—	Chosen
Tank inventory cap (case 1 / 2)	∞ 13.5 grams	Chosen / Update	
Updated variables	**Value**	**Unit**	**Reference**
Control segments N_t	20	—	Chosen

Table 5.4: Tank and delivery-agreement settings for the constant-delivery results (in addition to Table 5.1). N_t is refined to 20 segments (10-day resolution) for the tank scenarios.

5.3.2 Limited tank reserve via scheduled mode change

The unconstrained optimum in Figure 5.4 fills storage to roughly 0.27 kg—more than many practical tanks would hold. Figure 5.5 illustrates an alternative constructed from the same contract and κ_{sell}, targeting a **maximum inventory of about half** that peak (≈ 0.14 kg). The schedule has three phases:

1. **Gentle pot-iso** (U, T_{in} slightly below the $\kappa_{\mathrm{sell}} = \frac{1}{9}$ optimum) fills the tank more slowly and reduces early degradation stress.
2. **Galvanostatic window** (fixed U, stepped T_{in} tuned to track $\dot{n}_{\mathrm{H_2},\mathrm{agreement}}$) holds delivery while inventory approaches the cap.
3. **Closing pot-iso** (return to U^*, hold T_{in} at the galvanostatic endpoint) lets production track degradation while inventory stays bounded.

This mode change is not another degenerate NLP minimum; it is a forward replay shaped to respect a practical storage limit. Operationally it is preferable when inventory capacity is finite or carrying reserve hydrogen has off-model costs, even though the economic objective may differ slightly from Figure 5.4.

The accumulated revenue is ≈ 12.15 EUR and the terminal salvation term is ≈ 1.54 EUR. Peak inventory (≈ 0.14 kg) is about half the hoarding optimum in Figure 5.4 (≈ 0.27 kg), with no discontinuity in T_{in} at the phase boundaries.

5.3.3 Liquidation sensitivity

An additional sensitivity analysis has been made on the terminal liquidation multiplier κ_{sell} at the same agreement rate (6.5 NL/h) and pot-iso control mode. Figure 6.1 shows how optimal end-of-horizon tank inventory and net revenue respond as κ_{sell} is swept from 0 to 1. At low κ_{sell} the optimizer hoards hydrogen (Figure 5.4); above a breaking point near $\kappa_{\mathrm{sell}} \approx 33\%$ it becomes worthwhile to align production with the unrestricted baseline (Section 5.2). The choice $\kappa_{\mathrm{sell}} = \frac{1}{9}$ for the main results sits in the hoarding branch and exposes the tank-scheduling behaviour in Figure 5.4 and Figure 5.5. Further discussion is in Section 6.4.1.

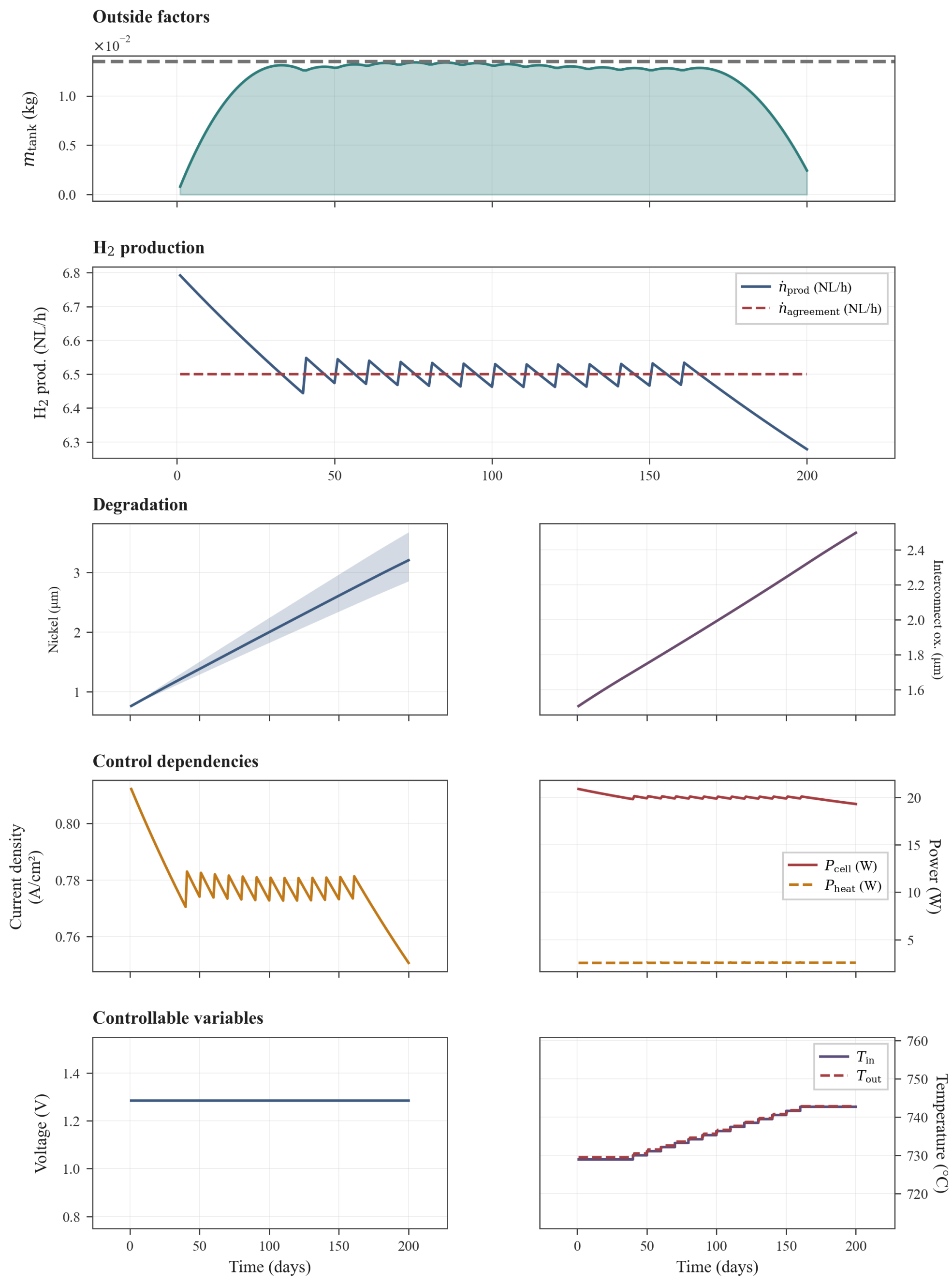


Figure 5.5: Half-capacity tank replay at $\kappa_{sell} = \frac{1}{9}$ and $\dot{n}_{H_2,agreement} = 6.5\,\mathrm{NL/h}$ (Table 5.4). Reduced pot-iso fill (segments 0–3; $U \approx 1.289\,\mathrm{V}$, $T_{in} \approx 735\,^{\circ}\mathrm{C}$), galvanostatic middle via stepped T_{in} at $U \approx 1.289\,\mathrm{V}$ (segments 4–10, ramp to $\approx 748\,^{\circ}\mathrm{C}$), and closing pot-iso hold at that same T_{in} (segments 11–19). Tank inventory peaks at $\approx 0.14\,\mathrm{kg}$ (day 110).

Setting	Value	Unit	Reference
Temperature continuity penalty w_T	10^{-3}	—	Chosen (Appendix A6)
Voltage continuity penalty w_U	10	—	Chosen (Appendix A6)
Updated variable	**Value**	**Unit**	**Reference**
Horizon T	24	h	Chosen
Control segments N_t	12	—	Chosen
Terminal liquidation κ_{sell}	0	—	Chosen
Initial tank inventory $m_{\text{tank},0}$	$m_{\text{tank},T}$	—	Periodic (4.1.4)
Power price $c_{P,k}$	$\bar{c}_P(1 - a\cos\theta_k)$, $\theta_k = 2\pi(h_k - 12\,\text{h})/(24\,\text{h})$	EUR/(W·s)	Chosen
Mean power price $\bar{c}_P$	0.079	EUR/kWh	[19]
Relative price swing a	0.9	—	Chosen

Table 5.5: Updated model settings for the 24-hour cyclic scenario. No terminal sale of hydrogen is allowed ($\kappa_{\text{sell}} = 0$), and the initial tank content is enforced as a periodic boundary. Twelve control segments are used for the day. The piecewise-constant power price on segment k follows a single cosine cycle over 24 h, with h_k the hour-of-day at the segment midpoint; the minimum $c_{P,k} = \bar{c}_P(1 - a)$ occurs at midday ($h_k = 12\,\text{h}$) and the maximum $\bar{c}_P(1 + a)$ at midnight, modelling lower prices when solar supply is high ($a = 0.9$ gives $\approx 19\times$ min/max spread). The continuity weights w_U and w_T enter the soft penalty Appendix A6, including the last→first segment wrap consistent with the repeating day; with potentiostatic voltage only w_T is effective.

5.4 24-hour cycle with varying power price

The final scenario shortens the horizon to one day and replaces the flat power price with a time-varying profile. The cosine form in Table 5.5 is a stylized stand-in for solar-driven day–night spreads: electricity is cheapest around midday and most expensive around midnight, with a relative swing $a = 0.9$ giving roughly a $19\times$ ratio between the midnight maximum and the midday minimum. Together with the contractual delivery setup from Section 5.3, this case tests whether the optimizer shifts production toward low-price periods while still meeting the agreement.

Terminal liquidation is disabled ($\kappa_{\text{sell}} = 0$), and the cyclic inventory constraint Equation 4.9 is enforced so that $m_{\text{tank}}(0) = m_{\text{tank}}(T)$. The day can therefore be studied as a repeating operating window without end-of-horizon sell-off. The updated settings are listed in Table 5.5.

Voltage is held potentiostatic and a large weight w_U discourages segment-to-segment voltage changes; temperature remains the primary free control. The optimal trajectory is shown in Figure 5.6. Production is highest when the power price is low and reduced when prices rise at night. The response is qualitatively as expected for load shifting, but the profile is simple: the optimizer modulates inlet temperature rather than exploiting both voltage and temperature.

The economic decomposition of the optimized run is as follows. Integrated running revenue over the 24,h window is 0.074 EUR and the terminal salvation term is 1.91 EUR, giving a total objective of 1.98 EUR. Where the addition salvation loss through the first day is about 0.0019 EUR, so the net economy from operation is effectively 0.072 EUR for the $20\,\text{cm}^2$ cell.

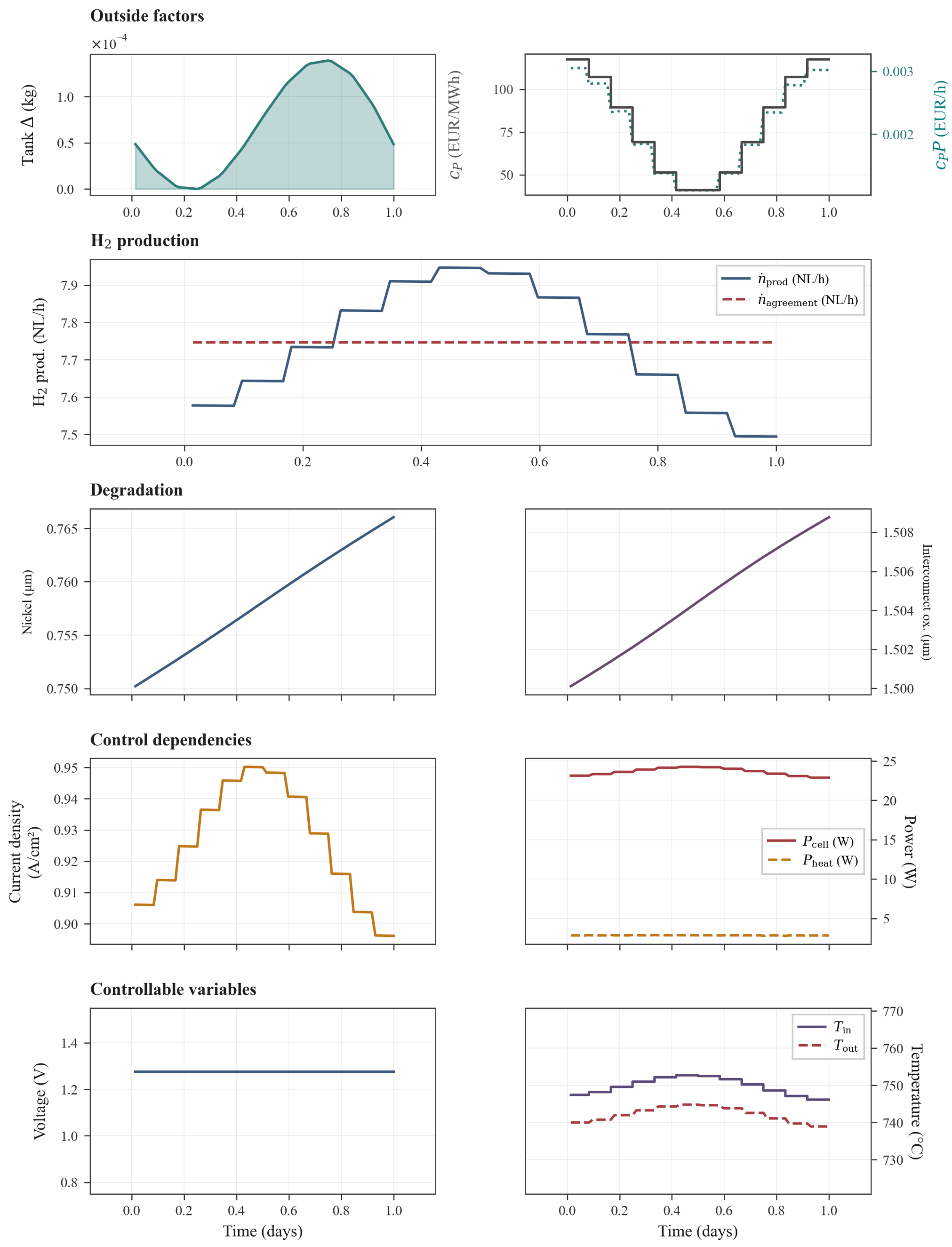


Figure 5.6: 24-hour cyclic optimum under a diurnal power price ($N_t = 12$; Table 5.5). *Figure pending re-run* with $a = 0.9$, periodic continuity wrap, and updated optimizer settings; values below are from an earlier $a = 0.5$ run. Fixed $U \approx 1.28$ V ($w_U = 10$); T_{in} is modulated ($w_T = 10^{-3}$) to shift production toward low-price midday segments (c_P and $c_P P$ in **Outside factors**). Peak $\dot{n}_{H_2,\text{prod}} \approx 7.95$ NL/h versus ≈ 7.5 NL/h overnight; the agreement rate is met on average via cyclic tank use ($\kappa_{\text{sell}} = 0$). Running revenue 0.07 EUR, terminal salvation 1.91 EUR.

Two modelling choices shape this outcome strongly. The cyclic tank constraint removes any terminal credit for leftover inventory and, together with the delivery obligation, forces production and storage to balance over the day rather than deferring output indefinitely. The continuity penalties bias the solution toward smooth temperature profiles across the full day, including the midnight wrap from the last segment back to the first, and through $w_U \gg w_T$ toward an effectively fixed voltage. In exploratory runs, increasing w_T pushes the optimizer back toward near-isothermal operation, whereas smaller values allow sharper concentration of production in the cheapest price windows.

5.4.1 Summary

This 24-hour case closes the results sequence begun in Section 5.2 and Section 5.3. Where the long-horizon baseline favours steady potentiostatic–isothermal operation and the delivery case exposes degenerate optima tied to tank use, the 24-hour cyclic run shows that time-varying electricity prices alone can induce deliberate scheduling—here through inlet temperature—while contractual delivery and periodic inventory are satisfied. Economically, almost all of the reported objective value sits in terminal salvation rather than the one-day cash flow (1.98 EUR total versus 0.07 EUR net operating economy), which is expected for a short horizon with little degradation.

The pattern is economically sensible but not surprising: shift electrolysis toward cheap power and lean on the tank to bridge expensive intervals. That simplicity is still informative, because it confirms that the revenue, storage, and price models are coupled as intended. At the same time, the result should not be overinterpreted: a single-harmonic price curve, fixed continuity weights, and the cyclic boundary are strong structural assumptions. Real markets exhibit ramps, negative prices, and correlation with renewable output that this stylized setup does not capture; the sensitivity to w_T, w_U, and the tank closure rule noted above underlines that operational conclusions from this scenario depend on those choices. A natural next step would be to test alternative terminal treatments, finer temporal resolution, or measured price series before drawing firm scheduling recommendations.

6 Discussion and Conclussion

6.1 Introduction

The results in Section 5.1–Section 5.4 suggest that potentiostatic-isothermal operation is preferred. The optimizer seeks to get as close to this operational state as possible when contracted delivery is introduced, adjusting only the temperature to avoid penalties for under-delivery. Similarly, in the daily varying price scenario, adjustments are made in temperature to shift production toward cheaper power intervals. In the unrestricted baseline (Section 5.2), the optimum is **exothermal** rather than thermoneutral: voltage sits above the thermoneutral value at the chosen inlet temperature, and nickel migration—not interconnect oxidation—reaches the $5\,\mu$m break limit at $t = T$. The salvation term and that hard cap together bound how aggressive this exothermal operation can be; without them, the optimizer would likely push further toward the temperature and current bounds. In the other two cases, control is dominated by the delivery contract and the price profile; the salvation term changes little over those horizons, so Ψ has limited influence on the optimal trajectory. But what does this mean? Section 6.2 and Section 6.3 interpret how the controls are used and whether the trajectories are plausible; Section 6.4 reviews the project-specific numerical defaults marked *Chosen* in the results tables; and Section 6.5 states which modelling and discretization choices bound how far those conclusions can be extrapolated. **??** collects extensions that follow from these findings.

6.2 Interpretation of optimal operating strategies

Taken together, the results indicate that extra control freedom is used only when contractual delivery or time-varying prices force a departure from potentiostatic-isothermal operation. The subsections below follow that logic.

6.2.1 Baseline without delivery obligations

The first question is how much of the available control freedom is actually used. For the unrestricted baseline (Section 5.2), the answer is very little: without a delivery floor, maximizing $\int R\,\mathrm{d}t + \Psi$ favours steady efficient production, and the optimal mode remains potentiostatic-isothermal. The additional decision variables are largely redundant in this scenario.

The baseline optimum is nevertheless **exothermal**: $U^* \approx 1.32\,\mathrm{V}$ exceeds the thermoneutral voltage at $T^*_{\mathrm{in}} \approx 751\,^\circ\mathrm{C}$, so heat release raises the temperature along the channel (roughly 751 to $780\,^\circ\mathrm{C}$ at $t = T$; Figure 5.3). Revenue is maximized up to the nickel-migration break limit at $5\,\mu$m; interconnect oxidation stays near $3.6\,\mu$m and is not life-limiting. Because the chosen voltage is effectively pinned at this exothermal operating point, U is also constant across the other long-horizon cases considered. A repeat study with voltage fixed potentiostatic and only one remaining degree of freedom, either piecewise galvanostatic current with inferred temperature, or piecewise isothermal T_{in} with inferred current, would likely reproduce the same qualitative conclusions while reducing the decision space.

6.2.2 Effect of a fixed delivery contract

A second question concerns how controls are parametrized within each segment. Here, both U and T_{in} are held constant on every control interval because the NLP is set up that way (Section 4.2.6). In the delivery case (Section 5.3), production often fluctuates around the agreement rate; in those intervals, holding galvanostatic current fixed would be the more natural mode. Allowing the optimizer to choose the operating mode—potentiostatic, galvanostatic, or isothermal, within each interval would be a direct extension. Even without that choice, the results still show where galvanostatic operation arises and how tank inventory mediates the switch.

6.2.3 Temperature jumps and implementability

Piecewise-constant optimal controls can imply large instantaneous changes in T_{in} between segments. Such jumps are difficult to implement on hardware and are not charged directly in the base formulation. In the 24-hour run, a squared continuity soft penalty was added retrospectively (Appendix A6, Appendix A6). A more physical alternative would be to include the auxiliary preheat power required to move from one inlet set-point to the next within a segment duration, or to model thermal cycling as an additional degradation driver.

6.3 Implied lifetime and proximity to break limits

Across the long-horizon runs, current densities are relatively high and the modelled degradation states approach the chosen break limits within 200 days (Table 5.1). In the baseline replay (Section 5.2), mean nickel-migration thickness reaches the $5\,\mu$m cap at $t = T$ while interconnect oxidation remains well below its $10\,\mu$m limit. That window is far shorter than stack lifetimes reported in the SOEC literature. Beyrami et al. predict lifetimes above $50{,}000\,$h at $0.6\,\mathrm{A\,cm^{-2}}$ when component failure criteria are enforced [2], and later embed stack lifetimes of 4–5 years into plant-scale economic optimization [3]. Giridhar et al. obtain replacement schedules on the order of 2–5 years depending on electricity price and operating mode [9].

The gap is partly expected: this thesis maximizes revenue plus salvation over a fixed 200-day horizon rather than full lifetime, and the delivery scenarios include galvanostatic periods that accelerate the modelled degradation mechanisms. It nevertheless suggests that one or more modelling choices may be too permissive. The fitted nickel-migration rate law (Section 3.2.2) is a simple empirical relation and may overpredict thickness growth at the currents seen in optimization.

6.3.1 Revenue–salvation balance in the objective

A complementary explanation lies in the scalar objective itself. The weighting between running revenue and terminal salvation may be misaligned: if c_{cell} or α_{Ψ} in Equation 4.4 understate the cost of approaching end-of-life, the optimizer can accept exothermal temperature and current profiles that literature-scale lifetime studies would reject. In the baseline, the monotonic heating along the channel homogenises nickel-migration growth (Figure 5.3), so failure is approached almost uniformly rather than at a single hot spot—but the $5\,\mu$m cap is still reached at $t = T$. Ohmic losses grow linearly with the resistances induced by nickel migration and interconnect oxidation, while the degradation rates themselves generally slow as the states evolve. A marginal increase in temperature therefore buys disproportionately more revenue for a given increment in resistance. The salvation term is bounded and does not rise sharply enough everywhere in state space to offset that trend unless α_Ψ is chosen very large.

One alternative is to replace the hard break limits used here with a failure probability that scales revenue at each instant. A schematic form is

$$\int_0^T \left(1 - P_{\text{fail}}(U, T, I, \mathbf{x})\right) R \,\mathrm{d}t + \mathbb{E}[\tau_{\text{life}} \mid \mathbf{x}_T]\,\mathbb{E}[R \mid \mathbf{x}], \tag{6.1}$$

where P_{fail} is the probability that the cell reaches a failure criterion given the controls and degradation state $\mathbf{x}$, and $\mathbb{E}[\tau_{\text{life}} \mid \mathbf{x}_0]$ is the expected remaining lifetime from the initial state.

Rather than stopping the simulation at a hard cap, revenue would be discounted by the instantaneous risk of failure, and the second term would reward expected production over life. In principle, P_{fail} could rise sharply once degradation states pass critical thresholds, giving a stronger, and nonlinear, penalty for operating too aggressively than the linear resistance increase combined with the bounded salvation term. Such a formulation would also ease the transition from cell-scale models to stack- and plant-scale objectives.

Such a model would require much more data than is currently available. However, an approximate model might be feasible with a comprehensive literature review on the probability of failure for each cell component, as this information is likely partially known. From there, it could be possible to construct a probabilistic model, though this would require substantial additional work. With such a model, it would then be possible to consider whether to optimize for expected revenue or for a different objective—such as a lower quantile of performance—depending on the desired risk tolerance.

6.4 Justification of project-specific settings

The results chapter marks several inputs as *Chosen* in Table 5.1, Table 5.4, and Table 5.5. Settings for the horizon, control segments, finite-volume mesh, operating bounds, and α_Ψ are already motivated in Section 5.1 and the grid-independence study (Section 4.2.3.2), which covers the baseline case (Table 5.1). The subsections below focus on the delivery and cyclic scenarios.

6.4.1 Delivery setup

Tank capacity is not listed in the tables because storage is modelled as unlimited. A finite capacity could be imposed, but its size would be another qualitative assumption; an uncapped tank isolates the delivery contract and penalty structure from inventory-cap effects.

The contractual delivery rate and the multipliers κ_{buy} and κ_{sell} are strongly coupled to the optimal control. Exploratory runs in Section 5.3 guide the reported values. If $\dot{n}_{\mathrm{H_2},\text{agreement}}$ is raised above the unconstrained

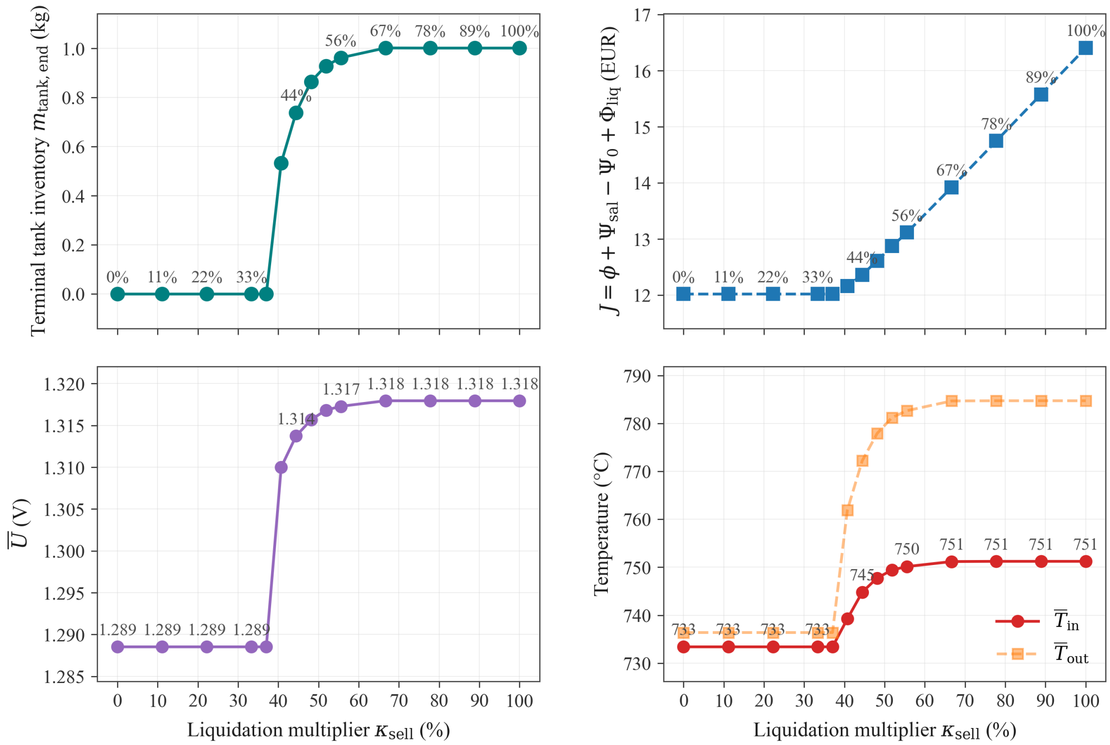


Figure 6.1: Sensitivity on the resale price from the leftover tank content at the end time. Here we see the tank content plottet at the left plot and the net revenue plottet on the right plot. The faded point at 55% is here because the optimizer failed to find an optimum for this value. Here it is clear to see that we find a breaking point for which the hydrogen price hits a low enough value that it is not worth producing. For a very small window it makes sense to produce less that the optimum to save the cell.

production rate at the baseline optimum (Section 5.2), the solver eventually drives the cell to the break limits, because sustaining that throughput is too aggressive. If the agreement rate is set below the baseline terminal production rate ($\approx 7.5\,\text{NL/h}$ at $t = T$ in Section 5.2), the solution depends mainly on κ_{sell}: with $\kappa_{\text{sell}} = 1$, the optimum coincides with the unrestricted baseline (including terminal tank liquidation); with $\kappa_{\text{sell}} = 0$, inlet temperature is reduced so that $m_{\text{tank}}(T) = 0$. Intermediate values of κ_{sell} interpolate between these extremes. The interpolation is shown in Figure 6.1, where it is clear that a breaking point exists near $\kappa_{\text{sell}} \approx 33\%$: below this value the optimizer hoards hydrogen in the tank (Figure 5.4); above it, production aligns with the unrestricted baseline (Section 5.2). Figure 5.5 illustrates how a constructed three-phase schedule can cap inventory at roughly half the hoarding optimum when storage is limited in practice.

The value $\kappa_{\text{sell}} = \frac{1}{9}$ used in Table 5.4 is intended as a rough market valuation: surplus hydrogen at the horizon is credited at roughly 11% of the contract price, reflecting competition with lower-cost *black* hydrogen (Section 5.3). The spot-purchase penalty $\kappa_{\text{buy}} = 1.5$ makes under-production costly enough that the optimizer avoids sustained shortfalls.

Together, the entries in Table 5.4 were selected because they expose informative tank behaviour at $\dot{n}_{H_2,\text{agreement}} = 6.5\,\text{NL/h}$: hoarding at low κ_{sell} (Figure 5.4) and a practical half-capacity schedule with galvanostatic tracking (Figure 5.5).

6.4.2 Cyclic setup

For the 24-hour case (Table 5.5), the horizon and N_t were scaled to one day with twelve two-hour segments. Terminal liquidation is set to $\kappa_{\text{sell}} = 0$ so that the periodic inventory constraint from Section 4.1.4 governs end-of-day inventory rather than a sell-off term.

The continuity weights were chosen so that a temperature step of roughly 5% of the bound window incurs a penalty on the order of 10% of the segment revenue; the voltage weight documents the potentiostatic setup (Section 5.4, Appendix A6). The cosine power-price profile provides a simple, interpretable day-night swing. Aside from the penalty weights, the cyclic parameters were not explored in a dedicated sensitivity study; qualitative scheduling patterns under those assumptions are reported in Section 5.4.

6.5 Model and numerical limitations

The results above are useful for qualitative comparison of operating modes, but several modelling and numerical choices bound how far the trajectories should be extrapolated. The subsections below separate physical scope from discretization and solver effects.

6.5.1 Physical scope and mechanism coverage

The stack model is a single one-dimensional channel with quasi-steady electrochemistry and temperature (Section 3.3, Table 3.1). Flow rates, pressure, and geometry are fixed; only U and T_{in} are optimized. That scale is appropriate for asking how a representative cell responds to price and delivery incentives, but it does not resolve stack-to-stack thermal coupling, manifold maldistribution, or plant-level auxiliary loads.

Degradation is limited to nickel migration and interconnect oxidation (Table 3.2). Other ageing paths—cathode delamination, chromium poisoning, seal leakage—are not modelled dynamically. Nickel agglomeration enters only as a fixed multiplier $\lambda_c \approx 0.55$ on the exchange current (Section 3.2.1), so any further cathode loss at high current is not captured. Together with the fitted nickel-migration rate law (Section 3.2.2), this narrow mechanism set likely contributes to the short implied lifetimes discussed in Section 6.3.

Several simplifying assumptions from Section 3.3 also matter at the currents and temperatures seen in optimization: ideal-gas behaviour, negligible mass-transport overpotential, and dropped axial heat conduction. IV validation against experiment (Figure 4.3) is good in the intended range, but error grows at high voltage; because the baseline optimum operates exothermally at $U^* \approx 1.32\,\text{V}$ with initial current density $\approx 1.35\,\text{A cm}^{-2}$, local model error may be larger than a near-thermoneutral replay would suggest.

6.5.2 Spatial and temporal discretization

Spatial resolution and the scenario-dependent values of N_t, T, and the continuity weights are discussed in Section 6.4. In brief, grid-independence analysis (Figure 4.4) indicates that the budget-limited mesh in Table 5.1 accepts spatial error on the order of 10^{-2}, which is adequate for comparing control modes but not for quantitative lifetime forecasting.

In time, controls are piecewise constant on N_t segments (Section 4.2.6) and degradation is advanced with implicit Euler on the same grid. The segment count varies between scenarios, so finer horizons do not automatically use finer control resolution.

6.5.3 Nonlinear programming and solution quality

The optimal control problem is transcribed as a single-shooting NLP solved with SciPy's trust-region method and adjoint gradients (Section 4.2.6, Appendix A5). Gradient checks support the implementation, but the problem is not guaranteed to be convex in practice. The delivery scenario (Section 5.3) returned two distinct local minima with equal objective value, which shows that reported trajectories need not be unique without extra constraints such as minimum inventory, continuity cost, or an operating-reserve preference.

Hard break limits on degradation states stop the forward integration when thresholds are reached. That keeps the solver well posed but introduces a non-smooth boundary that can interact with the bounded salvation term (Equation 4.4). Finally, the rolling horizon T is finite and scenario-dependent (200 days versus 24 hours), so terminal salvation can dominate short windows even when running revenue drives the qualitative scheduling pattern (Section 5.4).

6.6 Conclusions

1. **Control freedom is scenario-dependent.** Additional temporal freedom in U and T_{in} did not change the optimal operating mode for the unrestricted 200-day baseline: potentiostatic–isothermal operation at an **exothermal** set-point ($U^* \approx 1.32\,\text{V}$, $T_{\text{in}}^* \approx 751\,^\circ\text{C}$) remained optimal up to the nickel-migration break limit. Freedom became economically relevant only when a fixed hydrogen delivery contract and buffer tank were introduced.
2. **Delivery contracts activate inventory-aware operation.** With contractual offtake, the optimizer alternates between gentle potentiostatic–isothermal periods and galvanostatic periods dictated by tank inventory and penalty prices. Multiple control trajectories can achieve the same objective, so uniqueness should not be assumed without further constraints.
3. **Degradation-aware objectives favour operation bounded by nickel migration.** In the baseline, exothermal heating along the channel homogenises spatial nickel growth so the $5\,\mu$m cap is approached almost uniformly at $t = T$ (Figure 5.3), while interconnect oxidation remains secondary. The salvation term provides a consistent economic penalty as that limit is approached.
4. **The toolchain is fit for purpose at engineering fidelity.** IV validation, ODE replay against reference degradation data, and gradient checks support using the model for qualitative optimal-control studies, subject to the spatial-resolution and mechanism-scope limitations in Section 6.5.
5. **Objective choice shapes what "optimal" means.** The implemented form $\int_0^T R\,\mathrm{d}t + \Psi$ differs from the weighted schematic in Equation 2.1 and from LCOH- or efficiency-based objectives in the literature (Section 6.3). Several reported degeneracies—especially equivalent delivery trajectories—are artefacts of that scalar choice.

6.7 Use of generative AI tools

7 Future Work

The preceding chapters establish a cell-scale optimal-control toolchain and use it to compare baseline, contractual-delivery, and diurnal-price scenarios. The discussion in Section 6.1-Section 6.5 also highlights where the present model, objective, and solver stop short of plant-scale decision support. The items below are grouped by theme; they extend the work without repeating conclusions already drawn in **??**.

7.1 Modelling and degradation

- Extend the degradation state beyond nickel migration and interconnect oxidation, or calibrate the fitted rate laws against long-duration experiments at the current densities seen in optimization.
- Replace the fixed agglomeration multiplier λ_c with a dynamic state when operating far from the calibration point.
- Include auxiliary preheat power or a thermal-cycling degradation term so that segment-to-segment temperature changes carry an explicit cost (Section 6.2).
- Move from a single channel to stack- or plant-scale models with shared thermal and gas-management constraints.

7.2 Objectives and economics

- Implement the risk-aware objective sketched in Equation 6.1, or compare the present $\int R\,\mathrm{d}t + \Psi$ form directly with LCOH- and efficiency-based objectives from the literature (Section 6.3).
- Break objective degeneracies in the delivery case with a minimum-inventory penalty, reserve constraint, or preference for smoother tank utilisation.
- Test sensitivity to κ_{buy}, κ_{sell}, α_{Ψ}, and alternative terminal treatments for tank inventory (Section 5.4).

7.3 Control parametrization

- Repeat the scenario suite with voltage fixed potentiostatic and a single remaining degree of freedom, as suggested in Section 6.2.
- Allow the optimizer to select the operating mode per segment—potentiostatic, galvanostatic, or isothermal—rather than holding both U and T_{in} constant on every interval.
- Refine N_t and compare piecewise-constant schedules against measured day-ahead or intraday price series instead of the single-harmonic profile in Table 5.5.

7.4 AC:DC Optimization [17]

- Develop an objective function specifically designed for very short time horizon optimizations, in which the cell's mechanical and electrochemical responses cannot be assumed to be instantaneous. This would require accounting for the finite dynamic response of the cell, capturing any lag between input changes and their physical effects.
- Extend the degradation modelling framework to include not only irreversible but also reversible degradation mechanisms. By doing so, it becomes possible to study phenomena where the cell's performance may partially recover under certain operating conditions, enabling a more nuanced optimization of lifetime and efficiency.
- Formulate a single-period control trajectory that is completely unconstrained in shape, allowing the optimization procedure to discover non-standard or unexpected control profiles that might yield improved performance. This unconstrained formulation could help identify new strategies that would be otherwise overlooked under more restrictive parameterizations.
- Establish realistic constraints for the periodic control function in order to ensure that the resulting schedules are physically feasible and implementable in practice. These constraints could include bounds on ramp rates, limits on the magnitude of control variables, or conditions that enforce smooth transitions between periods, reflecting actual system limitations and safety requirements.

8 References

[1] Cost of hydrogen production | european hydrogen observatory. Retrieved from https://observatory.clean-hydrogen.europa.eu/hydrogen-landscape/production-trade-and-cost/cost-hydrogen-production?utm_source=chatgpt.com

[2] Javid Beyrami, Rafael Nogueira Nakashima, Arash Nemati, and Henrik Lund Frandsen. 2025. Lifetime and performance of solid oxide electrolysis stacks and systems under different operation modes and conditions. *International Journal of Hydrogen Energy* 102, (February 2025), 980–995. https://doi.org/10.1016/J.IJHYDENE.2025.01.028

[3] Javid Beyrami, Rafael Nogueira Nakashima, Arash Nemati, and Henrik Lund Frandsen. 2026. Economic optimization of modular solid oxide electrolysis system considering degradation and lifetime. *Energy Conversion and Management* 348, (January 2026), 120782. https://doi.org/10.1016/J.ENCONMAN.2025.120782

[4] Javid Beyrami, Rafael Nogueira Nakashima, Arash Nemati, and Henrik Lund Frandsen. 2024. Degradation modeling in solid oxide electrolysis systems: A comparative analysis of operation modes. *Energy Conversion and Management: X* 23, (2024), 100653. https://doi.org/https://doi.org/10.1016/j.ecmx.2024.100653

[5] Tomasz Brylewski, Sebastian Molin, Mirosław Stygar, Maciej Bik, Piotr Jeleń, Maciej Sitarz, Aleksander Gil, Ming Chen, and Peter Vang Hendriksen. 2024. Oxidation kinetics and electrical properties of oxide scales formed under exposure to air and ar–H2-H2O atmospheres on the crofer 22 h ferritic steel for high-temperature applications such as interconnects in solid oxide cell stacks. *International Journal of Hydrogen Energy* 94, (December 2024), 209–222. https://doi.org/10.1016/J.IJHYDENE.2024.11.091

[6] Qiong Cai, Claire S. Adjiman, and Nigel P. Brandon. 2014. Optimal control strategies for hydrogen production when coupling solid oxide electrolysers with intermittent renewable energies. *Journal of Power Sources* 268, (December 2014), 212–224. https://doi.org/10.1016/J.JPOWSOUR.2014.06.028

[7] Andrea Capolei. 2013. Numerical methods for oil reservoirs management. PhD thesis. Technical University of Denmark. Retrieved from https://people.compute.dtu.dk/jbjo/PhD_Thesis_2013_AndreaCapolei.pdf

[8] Robert Eymard, Thierry Gallouët, and Raphaèle Herbin. 2000. Finite volume methods. *Handbook of Numerical Analysis* 7, (January 2000), 713–1018. https://doi.org/10.1016/S1570-8659(00)07005-8

[9] Nishant V. Giridhar, Douglas A. Allan, Mingrui Li, Stephen E. Zitney, Lorenz T. Biegler, and Debangsu Bhattacharyya. 2024. Optimal operation of solid-oxide electrolysis cells considering long-term chemical degradation. *Energy Conversion and Management* 319, (November 2024), 118950. https://doi.org/10.1016/J.ENCONMAN.2024.118950

[10] Ernst Hairer and Gerhard Wanner. 1996. *Solving ordinary differential equations II: Stiff and differential-algebraic problems* (2nd ed.). Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-05221-7

[11] R. J.. Kee, Michael Elliott. Coltrin, Peter. Glarborg, and Huayang. Zhu. 2018. Chemically reacting flow : Theory, modeling, and simulation. (2018), 747.

[12] D. Larrain, J. Van herle, and D. Favrat. 2006. Simulation of SOFC stack and repeat elements including interconnect degradation and anode reoxidation risk. *Journal of Power Sources* 161, (October 2006), 392–403. https://doi.org/10.1016/J.JPOWSOUR.2006.04.151

[13] Andre Leonide, Yannick Apel, and Ellen Ivers-Tiffee. 2009. SOFC modeling and parameter identification by means of impedance spectroscopy. *ECS Transactions* 19, (October 2009), 81–109. https://doi.org/10.1149/1.3247567

[14] S. Primdahl and M. Mogensen. 1998. Gas conversion impedance: A test geometry effect in characterization of solid oxide fuel cell anodes. *Journal of The Electrochemical Society* 145, 7 (July 1998), 2431. https://doi.org/10.1149/1.1838654

[15] Tobias K. S. Ritschel, Andrea Capolei, Jozsef Gaspar, and John Bagterp Jørgensen. 2018. An algorithm for gradient-based dynamic optimization of UV flash processes. *Computers & Chemical Engineering* 114, (June 2018), 281–295. https://doi.org/10.1016/J.COMPCHEMENG.2017.10.007

[16] Omid Babaie Rizvandi, Matti Noponen, Henrik Lund Frandsen, and Xiufu Sun. 2025. Numerical investigation of electrochemical performance of commercial solid oxide cell stacks. *International Journal of Hydrogen Energy* 102, (February 2025), 830–844. https://doi.org/10.1016/J.IJHYDENE.2025.01.101

[17] Theis Løye Skafte, Omid Babaie Rizvandi, Anne Lyck Smitshuysen, Henrik Lund Frandsen, Jens Valdemar Thorvald Høgh, Anne Hauch, Søren Knudsen Kær, Samuel Simon Araya, Christopher Graves, Mogens Bjerg Mogensen, and Søren Højgaard Jensen. 2022. Electrothermally balanced operation of solid oxide electrolysis cells. *Journal of Power Sources* 523, (2022), 231040. https://doi.org/10.1016/j.jpowsour.2022.231040

[18] Bengt Sundén. 2019. Electrochemistry and thermodynamics. *Hydrogen, Batteries and Fuel Cells* (2019), 15–36. https://doi.org/10.1016/B978-0-12-816950-6.00002-6

[19] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. 2020. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* 17, (2020), 261–272. https://doi.org/10.1038/s41592-019-0686-2

[20] Wikipedia contributors. 2025. Projectile motion — Wikipedia, the free encyclopedia.

A1 Deriving PDEs for species balances

Following [14], eqs. [A-3]–[A-5]:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = \underbrace{A_{int}N_{in,\mathrm{H_2O}}}_{\text{Inlet}} + \underbrace{A_{cell}N_{a,\mathrm{H_2O}}}_{\text{Cell contribution}} - \underbrace{A_{int}N_{out,\mathrm{H_2O}}}_{\text{Outlet}}$$

$$N_{in/out,\mathrm{H_2O}} = v_{fuel}\,\mu_{in/out,\mathrm{H_2O}}$$

$$N_{a,\mathrm{H_2O}} = \nu_{\mathrm{H_2O}}\frac{i}{2F}$$

The equations are also required in terms of the concentration $\mu_{\mathrm{H_2O}}$, defined by:

$$\mu_{\mathrm{H_2O}} = \frac{n_{\mathrm{H_2O}}}{V}$$

Substituting these relations and the ideal gas law, then discretizing, gives

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{A_{cell}}{V}\nu_{\mathrm{H_2O}}\frac{i}{2F} + \frac{A_{int}}{V}v_{fuel}(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))$$

With $A_{cell} = w\Delta z$ and $V = \Delta z A_{int}$, isolating the time derivative yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w\Delta z}{A_{int}\Delta z}\nu_{\mathrm{H_2O}}\frac{i}{2F} + \frac{A_{int}}{A_{int}\Delta z}v_{fuel}(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))$$

$$= \frac{w}{A_{int}}\nu_{\mathrm{H_2O}}\frac{i}{2F} + v_{fuel}\frac{(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))}{\Delta z}$$

Letting $\Delta z \to 0$ yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}\nu_{\mathrm{H_2O}}\frac{i}{2F} - v_{fuel}\frac{\partial \mu_{\mathrm{H_2O}}}{\partial z}$$

or, since $N_{H_2O} = v_{fuel}\mu_{H_2O}$,

$$\frac{\partial \mu_{H_2O}}{\partial t} = \frac{w}{A_{int}} \nu_{H_2O} \frac{i}{2F} - \frac{\partial N_{H_2O}}{\partial z}$$

The same steps for H_2 and O_2 give

$$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}} \nu_i \frac{i}{2F} - \frac{\partial N_i}{\partial z}, \qquad \text{for } i \in \{H_2O, H_2, O_2\}$$

with $\nu_i = \left(1, -1, -\frac{1}{2}\right)^\top$.

A1.1 Unit check (H_2O)

A unit check on the H_2O equation gives

$$\frac{\partial \mu_{H_2O}}{\partial t} = \frac{w}{A_{int}} \frac{i}{2F} - v_{fuel} \frac{\partial \mu_{H_2O}}{\partial z}$$

Left side units: - $\frac{\partial \mu_{H_2O}}{\partial t}$: $[mol \cdot m^{-3} \cdot s^{-1}]$

Right side units: - $\nu_{H_2O} \frac{w}{A_{int}} \frac{i}{2F}$: - ν_{H_2O} is dimensionless. - $\frac{w}{A_{int}}$: $[m]/[m^2] = [m^{-1}]$ - $\frac{i}{2F}$: $[A \cdot m^{-2}]/[C \cdot mol^{-1}]$

$$[m^{-1}] \cdot [A\, m^{-2}]/[C\, mol^{-1}] = [A\, m^{-3}]/[C\, mol^{-1}]$$

Next, since $1\, A = 1\, C/s$:

$$= [C\, s^{-1}\, m^{-3}]/[C\, mol^{-1}] = [mol\, s^{-1}\, m^{-3}]$$

So the total units are $[mol\, m^{-3}\, s^{-1}]$

- $v_{fuel} \frac{\partial \mu_{H_2O}}{\partial z}$: $[m \cdot s^{-1}] \cdot \frac{[mol \cdot m^{-3}]}{[m]} = [mol \cdot m^{-3} \cdot s^{-1}]$ (since $\partial \mu / \partial z$ has units concentration per length, i.e. $[mol \cdot m^{-3}]/[m] = [mol \cdot m^{-4}]$).

Both terms on the right match the left-hand side, $[mol \cdot m^{-3} \cdot s^{-1}]$.

A2 Deriving PDEs for energy balances

Starting from the thermal energy equation with $\mathcal{S} = \{H_2O, O_2, H_2\}$:

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T$$

With $\frac{\partial p}{\partial t} = \frac{\partial p}{\partial z} = 0$ and $h \approx c_p T$, this becomes

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (c_p T N_i) + Q_T$$

Neglecting thermal conductivity,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \nabla \cdot (T N_i) + Q_T$$

In one dimension,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \frac{\partial T N_i}{\partial z} + Q_T$$

With $N_i = v_i \mu_i$,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + Q_T$$

A2.1 Source term

The volumetric source follows from the cell power,

$$P = i A_{cell} U,$$

which is the current times the cell voltage. Dividing by the volume gives Q_T:

$$Q_T = \frac{P}{V} = \frac{A_{cell}}{V} iU$$

With $A_{cell} = wdz$ and $V = A_{int}dz$,

$$Q_T = \frac{w}{A_{int}} iU$$

The full equation is therefore

$$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i\in\mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + \frac{w}{A_{int}} iU$$

A2.2 Steady-state limit

Expanding the advective term and setting $\partial_t T = 0$ gives

$$0 = -\sum_{i\in\mathcal{S}} v_i c_{p,i} \left(\frac{\partial T}{\partial z} \mu_i + T \frac{\partial \mu_i}{\partial z} \right) + \frac{w}{A_{int}} iU$$

Substituting $h_i = c_{p,i}T$ and the steady-state species balance $v_i \frac{\partial \mu_i}{\partial z} = \frac{w}{A_{int}} \nu_i \frac{i}{n_e F}$ yields

$$0 = -\sum_{i\in\mathcal{S}} \left(v_i c_{p,i} \frac{\partial T}{\partial z} \mu_i + \frac{w}{A_{int}} \frac{i}{n_e F} \nu_i h_i \right) + \frac{w}{A_{int}} iU$$

Factoring out the common terms,

$$0 = -\frac{\partial T}{\partial z} \sum_{i\in\mathcal{S}} v_i c_{p,i} \mu_i - i \frac{w}{A_{int}} \sum_{i\in\mathcal{S}} \nu_i h_i + i \frac{w}{A_{int}} U$$

With

$$\sum_{i\in S} v_i c_{p,i} \mu_i = C_p, \quad U_{th} = \frac{1}{n_e F} \sum_{i\in\mathcal{S}} \nu_i h_i$$

this gives

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z}$$

A3 Computing the finite-volume discretization

A3.1 Species balances

From Equation 3.5, Equation 3.6 and Equation 3.7, integration over a cell volume gives

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \int_{\Omega_k} \frac{w}{A_{int}} N_{c,i}\, dV - \int_{\Omega_k} \frac{\partial N_i}{\partial z}\, dV$$

The time derivative can be taken outside the integral, and integrating the concentration over a volume gives the moles in that volume:

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \partial_t \int_{\Omega_k} \mu_i\, dV = \frac{\partial n_i}{\partial t} = \dot{n}_i$$

Using the approximation in Section 4.2.2 with $V_k = A_{int}\Delta z$,

$$\int_{\Omega_k} \frac{w}{A_{int}} N_{c,i}\, dV \approx \frac{w}{A_{int}} N_{c,i}(z_k) V_k = w N_{c,i}(z_k)\Delta z$$

For the flux term, integration over z is performed first, then over the cross-sectional area:

$$\begin{aligned}\int_{A_{int}} \int_{z_{k-\frac{1}{2}}}^{z_{k+\frac{1}{2}}} \frac{\partial N_i}{\partial z}\, dz\, dA &\approx \int_{A_{int}} \Big(N_i(z_k) - N_i(z_{k-1}) \Big)\, dA \\ &= \Big(N_i(z_k) - N_i(z_{k-1}) \Big) A_{int}\end{aligned}$$

Each cell therefore satisfies

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) = \dot{n}_i(z_k) \approx w N_{c,i}\Delta z - A_{int}(N_i(z_k) - N_i(z_{k-1}))$$

A3.2 Energy balance

The steady-state form is discretized first, then the transient term.

A3.2.1 Steady state

Starting from Equation 3.18:

$$0 = i\frac{w}{A_{int}}(U - U_{th}) - C_p\frac{\partial T}{\partial z}$$

The same procedure as for the species balances gives

$$0 = \int_{\Omega_k} i\frac{w}{A_{int}}(U - U_{th}) - \int_{\Omega_k} C_p\frac{\partial T}{\partial z}$$

As before,

$$\int_{\Omega_k} C_p\frac{\partial T}{\partial z}\,dV \approx C_p(T(z_k) - T(z_{k-1}))A_{int}$$

Collecting terms,

$$0 \approx iw(U - U_{th})\Delta z - C_p(T(z_k) - T(z_{k-1}))A_{int}$$

A3.2.2 Transient term

For the time derivative,

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} = \int_{\Omega_k} Q_T - v\sum_{i\in\mathcal{S}}\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z}$$

On the left-hand side, ρ integrated over the cell volume is the cell mass m_k, so

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} \approx m_k c_p\frac{\partial T}{\partial t}(z_k)$$

The source term follows the same pattern as above, $\int_{\Omega_k} Q_T = Q_T(z_k)A_{int}\Delta z$. For the advective flux, one species is shown explicitly; the others are analogous:

$$\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z} \approx c_{p,i}\Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big)A_{int}$$

The discrete energy balance is therefore

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T(z_k) A_{int} \Delta z - \sum_{i \in \mathcal{S}} v_i c_{p,i} \Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1}) \Big) A_{int}$$

A4 Relevant gradients for the numerical solver

A4.1 Molar balance

$$\text{Jacobian}_\mu[x] = \begin{bmatrix} \frac{1}{\mu^f} & -\frac{1}{\mu^f} & 0 \\ -\frac{1}{\mu^f} & \frac{1}{\mu^f} & 0 \\ 0 & 0 & \frac{\mu_{in,\mathrm{N_2}}}{(\mu_{in,\mathrm{N_2}}+\mu_{\mathrm{O_2}})} \end{bmatrix}$$

Then

$$\nabla_\mu f(x) = \left(\text{Jacobian}_\mu[x]\right)^\top \nabla_x f$$

A4.1.1 Nernst Equation

A4.1.1.1 Gradient for molar fractions: ∇_x

$$\nabla_x \eta_{nernst} = \frac{RT}{2F} \begin{bmatrix} -\frac{1}{x_{\mathrm{H_2O}}} \\ \frac{1}{x_{\mathrm{H_2}}} \\ \frac{1}{2x_{\mathrm{O_2}}} \end{bmatrix}$$

A4.1.1.2 Gradient for temperature: ∇_T

$$\nabla_T \eta_{nernst} = \frac{\nabla_T(\Delta G)}{2F} + \frac{R}{2F} \log \left\{ \frac{x_{\mathrm{H_2}}\sqrt{x_{\mathrm{H_2}}}}{x_{\mathrm{H_2O}}\sqrt{x_0}} \right\}$$

$$\nabla_T(\Delta G) = -0.0062T + -49.119$$

A4.1.2 Activation overpotential

A4.1.2.1 Gradient for molar fractions: ∇_x

$$\nabla_x(\text{Butler-Volmer}_\zeta) = \nabla_x(i_0^\zeta)\Big(\exp\big[\alpha_\zeta \tfrac{n_e F}{RT} \eta_{act}^\zeta\big] - \exp\big[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\big]\Big)$$

$$\nabla_x(i_0^a) = \gamma_a T \exp\left\{-\frac{E_{act}^a}{RT}\right\} \begin{bmatrix} 0 \\ 0 \\ m_{O_2}\left(\frac{x_{O_2}}{x_0}\right)^{m_{O_2}-1} \end{bmatrix}$$

$$\nabla_x(i_0^f) = \lambda_f \gamma_f T \exp\left\{-\frac{E_{act}^f}{RT}\right\} \begin{bmatrix} m_{H_2}\left(\frac{x_{H_2O}}{x_0}\right)^{m_{H_2O}-1}\left(\frac{x_{H_2}}{x_0}\right)^{m_{H_2}} \\ m_{H_2O}\left(\frac{x_{H_2O}}{x_0}\right)^{m_{H_2O}}\left(\frac{x_{H_2}}{x_0}\right)^{m_{H_2}-1} \\ 0 \end{bmatrix}$$

A4.1.2.2 Gradient for temperature: ∇_T

$$\begin{aligned}\nabla_T(\text{Butler-Volmer}_\zeta) &= \nabla_T(i_0^\zeta)\Big(\exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\right] - \exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\right]\Big) \\ &+ i_0^\zeta\Big(-\alpha_\zeta \tfrac{n_e F}{RT^2}\eta_{act}^\zeta \exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\right] - (1-\alpha_\zeta)\tfrac{n_e F}{RT^2}\eta_{act}^\zeta \exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\right]\Big)\end{aligned}$$

$$\nabla_T(i_0^\zeta) = \left(1 - \frac{E_{act}^\zeta}{RT}\right)\lambda_\zeta \gamma_\zeta \exp\left\{-\frac{E_{act}^\zeta}{RT}\right\}\prod_{i\in\mathcal{S}}\left(\frac{x_i}{x_0}\right)^{m_i}$$

A4.1.2.3 Gradient for activation overpotential: $\nabla_{\eta_{act}^\zeta}$

$$\nabla_{\eta_{act}^\zeta}(\text{Butler-Volmer}_\zeta) = i_0^\zeta\Big(\alpha_\zeta \tfrac{n_e F}{RT}\exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\right] + (1-\alpha_\zeta)\tfrac{n_e F}{RT}\exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\right]\Big)$$

A4.2 Degradation

A4.2.1 Ohmic degradation

A4.2.1.1 Gradient for molar fractions: ∇_x

$$\nabla_x R_{ohm} = \mathbf{0}$$

A4.2.1.2 Gradient for temperature: ∇_T

$$\nabla_T R_{ohm} = \left(1 - \frac{E_{a,ohm}}{RT}\right)\frac{1}{B_{ohm}}\exp\left\{\frac{E_{a,ohm}}{RT}\right\} + \sum_{d\in\mathcal{D}_{ohm}}\nabla_T R_d$$

$$\nabla_T R_{\text{Ni-mi}} = d_{\text{Ni-mi}}\frac{T - E_{\text{act,elec}}/R}{\sigma_{\text{elec,o}} T \exp\{-\frac{E_{\text{act,elec}}}{RT}\}}$$

A4.3 Temperature equations

A4.3.1 Specific heat

A4.3.1.1 Gradient for molar fractions: ∇_μ

$$\nabla_\mu C_p = \begin{bmatrix} c_{p,\mathrm{H_2O}} \\ c_{p,\mathrm{H_2}} \\ c_{p,\mathrm{O_2}} \end{bmatrix}$$

A4.3.1.2 Gradient for temperature: ∇_T

$$\nabla_T C_p = \sum_{i \in \mathcal{S}} \mu_i \nabla_T c_{p,i}$$

A4.3.2 Thermaneutral voltage

A4.3.2.1 Gradient for molar fractions: ∇_μ

$$\nabla_\mu U_{th} = \mathbf{0}$$

A4.3.2.2 Gradient for temperature: ∇_T

$$\nabla_T U_{th} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i \nabla_T h_i$$

A5 NLP Gradients

This appendix collects the gradients used in the NLP formulation of Equation 4.1. The discrete chain rule for the stage model is stated first, followed by the partial derivatives that enter it and the linear systems for sensitivities of the spatial FVM model and the slow trajectory in time.

A5.1 Discrete chain rule for the stage model

Between shooting stages, the integrated objective contribution ϕ and the degradation variables d satisfy a discrete sensitivity chain. For a scalar stage index i and a control component u_k,

$$\frac{\mathrm{d}}{\mathrm{d}u_k}\phi_{i+1} = \nabla_{u_i}\dot{\phi} \cdot \nabla_{u_k} u_i + \nabla_{d_i}\dot{\phi}\,\frac{\mathrm{d}}{\mathrm{d}u_k} d_i + \frac{\mathrm{d}}{\mathrm{d}u_k}\phi_i,$$

with the analogous relation for the degradation state,

$$\frac{\mathrm{d}}{\mathrm{d}u_k} d_{i+1} = (\nabla_{u_i}\dot{d}_i)\,\nabla_{u_k} u_i + (\nabla_{d_i}\dot{d}_i)\,\frac{\mathrm{d}}{\mathrm{d}u_k} d_i + \cdots,$$

where the trailing terms collect any additional dependence of the stage map on u_k (for example through implicit stage equations).

The partial gradients of the instantaneous rates are

$$\begin{aligned}
\nabla_u\dot{\phi} &= c_{\mathrm{H}_2}\nabla_u\frac{\partial n_{\mathrm{H}_2}}{\partial t} - c_P(u\nabla_u I - I),\\
\nabla_d\dot{\phi} &= c_{\mathrm{H}_2}\nabla_d\frac{\partial n_{\mathrm{H}_2}}{\partial t} - c_P u\nabla_d I - \left[c_{i,\text{utilization}}\frac{1}{d_{i,\text{critical}}}\right]_{i\in\mathcal{D}_{\text{fail}}},\\
\nabla_\phi\dot{\phi} &= 0,\\
\nabla_u\dot{d} &= c_2\nabla_u i + c_3\nabla_u T + c_4(\nabla_u i\,T + i\nabla_u T),\\
\nabla_d\dot{d} &= c_2\nabla_d i + c_3\nabla_d T + c_4(\nabla_d i\,T + i\nabla_d T),\\
\nabla_\phi\dot{d} &= 0.
\end{aligned}$$

Note that $\nabla_\zeta I = \frac{A_{\text{cell}}}{N_{\text{cell}}}\sum_{j=1}^{N_{\text{cell}}}\nabla_\zeta i_j$ for $\zeta \in \{u, d, \phi, \dots\}$.

The remaining ingredients are the gradients of the current density with respect to cell voltage and local degradation state; these are given below. Further electrochemical partials are listed in Appendix A4.

A5.2 Gradient with respect to cell voltage: $\nabla_u i$

The current is expressed in terms of fuel- and air-side activation branches in the same notation as Appendix A4. With scalars $\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)$ and $\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)$ denoting $\partial i/\partial\eta^f_{\text{act}}$ and $\partial i/\partial\eta^a_{\text{act}}$ at the operating point,

$$\nabla_u i = \frac{1}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

A5.3 Gradient with respect to degradation: $\nabla_d i$

Similarly, with $\nabla_d R_{\text{ohm}}$ the partial of the ohmic resistance with respect to the local degradation variable,

$$\nabla_d i = -i\,\frac{\nabla_d R_{\text{ohm}}}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

A5.4 Sensitivity of the spatial FVM discretization

At each time level the converged spatial state satisfies a residual of flux form. With cell index k along the stack, let $Q(u_k)$ denote the cellwise source residual and $F(u_k)$ the numerical flux between neighbours. The source term is scaled by the cell thickness Δz and the flux difference by the interface area A_{int}, so the semi-discrete steady relation is

$$0 = \Delta z\, Q(u_k) - A_{\text{int}}\left(F(u_k) - F(u_{k-1})\right).$$

Differentiating with respect to a parameter p (a control coefficient, a boundary value, etc.) and writing $\partial u_k/\partial p$ for the sensitivity of the converged state gives the familiar block recursion

$$\begin{aligned}\left(\Delta z\,\frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}}\frac{\partial F}{\partial u_k}(u_k)\right)\frac{\partial u_k}{\partial p} + A_{\text{int}}\frac{\partial F}{\partial u_{k-1}}(u_{k-1})\frac{\partial u_{k-1}}{\partial p}\\ = A_{\text{int}}\left(\frac{\partial F}{\partial p}(u_k) - \frac{\partial F}{\partial p}(u_{k-1})\right) - \Delta z\,\frac{\partial Q}{\partial p}(u_k).\end{aligned}$$

Stacking all cells, this is the linear system

$$J_{\text{fvm}}\frac{\partial \mathbf{u}}{\partial p} = b^{(p)}_{\text{fvm}},$$

with the block-lower-bidiagonal Jacobian and right-hand side

$$J_{\text{fvm}} = \begin{bmatrix} \Delta_1 & & & \\ A_{\text{int}} \dfrac{\partial F}{\partial u_1}(u_1) & \Delta_2 & & \\ & \ddots & \ddots & \\ & & A_{\text{int}} \dfrac{\partial F}{\partial u_{N-1}}(u_{N-1}) & \Delta_N \end{bmatrix},$$

$$\text{with } \Delta_k = \Delta z \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k),$$

$$b_{\text{fvm}}^{(p)} = \begin{bmatrix} A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_1) - \frac{\partial F}{\partial p}(u_{\text{in}}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_1) \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_2) - \frac{\partial F}{\partial p}(u_1) \right) - \Delta z \frac{\partial Q}{\partial p}(u_2) \\ \vdots \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_N) - \frac{\partial F}{\partial p}(u_{N-1}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_N) \end{bmatrix}$$

The same block structure matches the flux assembly used in Appendix A3.

A5.5 Sensitivity of the ODE system

For this section the problem is written in general form:

$$\begin{aligned} \max_u \Big\{ &\text{Objective} = \int_0^T \Phi(X, d, u)\, dt + \Psi(X, d, u) \Big\} \\ \text{s.t.} \quad & 0 = R_{\text{fvm}}(X, d, u) \\ & \dot{d} = f(X, d, u) \end{aligned}$$

The method of adjoint gradients [15], [7] is used, with the following discretization ($t_0 = 0 < t_1 \cdots t_i \cdots t_N = T$):

$$X(t_i) \approx X_i, \quad d(t_i) \approx d_i, \quad u(t_i) \approx u_i.$$

$$\begin{aligned} R_{\text{fvm},i} &= R_{\text{fvm}}(X_{i+1}, d_{i+1}, u_i), \\ D_i &= d_{i+1} - f(X_{i+1}, d_{i+1}, u_i)\Delta t - d_i \\ \Phi_i &= \Phi(X_{i+1}, d_{i+1}, u_i)\Delta t \approx \int_{t_i}^{t_{i+1}} \Phi(X, d, u) dt \end{aligned}$$

The combined residual for the FVM constraints and the ODE defect is then:

$$R_i = \begin{bmatrix} D_i \\ R_{\text{fvm},i} \end{bmatrix}$$

The state variables are combined as

$$w_i = \begin{bmatrix} X_i \\ d_i \end{bmatrix}$$

The Jacobian of the residual with respect to the internal and control variables is

$$\frac{\partial R_i}{\partial w_i} = \begin{bmatrix} \mathbf{0} & \frac{\partial D_i}{\partial d_i} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \frac{\partial R_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial D_i}{\partial X_{i+1}} & \frac{\partial D_i}{\partial d_{i+1}} \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial R_i}{\partial u_i} = \begin{bmatrix} \frac{\partial D_i}{\partial u_i} \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i} \end{bmatrix}$$

$$\frac{\partial \Phi_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial \Phi_i}{\partial X_{i+1}} \\ \frac{\partial \Phi_i}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial \Phi_i}{\partial u_i}$$

For implementation these blocks are spelled out in more detail:

$$\begin{aligned}
\frac{\partial R_i}{\partial w_i} &= \begin{bmatrix} \mathbf{0} & -\mathbf{I} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \\
\frac{\partial R_i}{\partial w_{i+1}} &= \begin{bmatrix} -\Delta t \frac{\partial f}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \mathbf{I} - \Delta t \frac{\partial f}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix} \\
\frac{\partial R_i}{\partial u_i} &= \begin{bmatrix} -\Delta t \frac{\partial f}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix}
\end{aligned}$$

The blocks $\partial R_{\text{fvm},i}/\partial X_{i+1}$ coincide with J_{fvm} from the spatial sensitivity section (same flux assembly). At each time level:

$$\frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} = J_{\text{fvm}} \quad \frac{\partial R_{\text{fvm},i}}{\partial u_i} = \begin{bmatrix} | & | \\ b_{\text{fvm}}^{(u_1^{(U)})} & b_{\text{fvm}}^{(u_i^{(T_0)})} \\ | & | \end{bmatrix} \quad \frac{\partial R_{\text{fvm},i}}{\partial d_i} = b_{\text{fvm}}^{(d)}$$

With these blocks, the objective gradient with respect to u_i is

$$\nabla_{u_i} \text{Objective} = \frac{\partial \Phi_i}{\partial u_i} + \left(\frac{\partial R_i}{\partial u_i}\right)^\top \lambda_{i+1} \quad \text{for} \quad i = 1, \dots, N$$

The adjoint variables $\Lambda = (\lambda_k)_k$ satisfy

$$\left(\frac{\partial R_{N-1}}{\partial w_N}\right)^\top \lambda_N = -\frac{\partial \Phi_{N-1}}{\partial w_N} - \frac{\partial \Psi}{\partial w_N},$$

$$\left(\frac{\partial R_{i-1}}{\partial w_i}\right)^\top \lambda_i + \left(\frac{\partial R_i}{\partial w_i}\right)^\top \lambda_{i+1} = -\frac{\partial \Phi_{i-1}}{\partial w_i}, \qquad i = 1, \dots, N-1.$$

Equivalently, one large block system can be assembled for Λ.

$$\begin{bmatrix} \frac{\partial R_0}{\partial w_1} & & & & \\ \frac{\partial R_1}{\partial w_1} & \frac{\partial R_1}{\partial w_2} & & & \\ & \ddots & \ddots & & \\ & & & \frac{\partial R_{N-2}}{\partial w_{N-1}} & \\ & & & \frac{\partial R_{N-1}}{\partial w_{N-1}} & \frac{\partial R_{N-1}}{\partial w_N} \end{bmatrix}^\top \Lambda = - \begin{bmatrix} \frac{\partial \Phi_0}{\partial w_1} \\ \frac{\partial \Phi_1}{\partial w_2} \\ \vdots \\ \frac{\partial \Phi_{N-2}}{\partial w_{N-1}} \\ \frac{\partial \Phi_{N-1}}{\partial w_N} + \frac{\partial \Psi}{\partial w_N} \end{bmatrix}$$

A6 Squared continuity penalty

Piecewise-constant controls on N_t segments can introduce jumps between adjacent intervals. When both voltage and inlet temperature are optimized segment-wise, the optimizer may exploit those jumps unless they are penalized. The implementation adds an optional soft penalty to the running objective, proportional to the sum of squared differences between neighbouring segment values.

For segment index $k = 0, \ldots, N_t - 2$, let U_k and $T_{\text{in},k}$ denote the piecewise-constant controls on segment k. With non-negative weights w_U and w_T (CLI: `--continuity-weight-u`, `--continuity-weight-T`), the penalty is

$$P_{\text{cont}} = w_U \sum_{k=0}^{N_t-2} (U_{k+1} - U_k)^2 + w_T \sum_{k=0}^{N_t-2} (T_{\text{in},k+1} - T_{\text{in},k})^2.$$

It is added to the running economic cost before integration over the horizon. Setting $w_U = w_T = 0$ disables the term (the default in Table 5.1).

In **potentiostatic** operation ($U_k \equiv U$), voltage is a single decision variable rather than a length-N_t profile, so adjacent voltage differences vanish and the w_U term contributes nothing. Only the inlet-temperature sum is active in that mode.

The penalty is smooth and cheap to evaluate; its gradient with respect to the control knots is implemented analytically in the NLP (`code/src/scripts/nlp/continuity.py`). Saved run metadata reports P_{cont}, P_U, and $P_{T_{\text{in}}}$ as the two sums in Appendix A6.

A7 Gradient verification plots

The NLP solver uses adjoint gradients assembled from the partial derivatives in Appendix A4 and Appendix A5. Before running the optimization cases in Section 5.1–Section 5.4, analytical derivatives were checked against central finite-difference (FD) approximations. Standalone scripts in `code/src/standalone/` generate the figures below; additional component-level checks exist in `code/figures/testers/grad` but are omitted here.

A7.1 End-to-end NLP objective

Figure A7.1 sweeps uniform cell voltage and inlet temperature on a single-shooting problem with $N_t = 100$ time segments ($\Delta t = 4320\,\mathrm{s}$), exercising the full adjoint chain over a long horizon. The adjoint totals $\sum_j \partial f/\partial U_j$ and $\sum_j \partial f/\partial T_{\mathrm{in},j}$ match the FD curves over the operating range.

A7.2 Spatial FVM Jacobian

The steady-state cell residual $r(\mu_{H_2}, \mu_{O_2}, T)$ must be differentiated when assembling the block-bidiagonal system in Appendix A5. Figure A7.2 compares each Jacobian entry to a secant FD estimate; the curves agree on the test grid.

A7.3 Electrochemical coupling

Current density enters both the degradation rates and the economic objective. Figure A7.3 and Figure A7.4 verify $\nabla_U i$ and $\nabla_d i$ from Appendix A5. Figure A7.5 checks the fuel-side Butler–Volmer branch and its partials w.r.t. temperature, composition, and activation overpotential.

A7.4 Instantaneous economic rate

Figure A7.6 checks partial derivatives of the running economic rate $\dot{\phi}$ w.r.t. outlet chemical potentials and the stack-outlet temperature that feed the stage objective in Appendix A5.

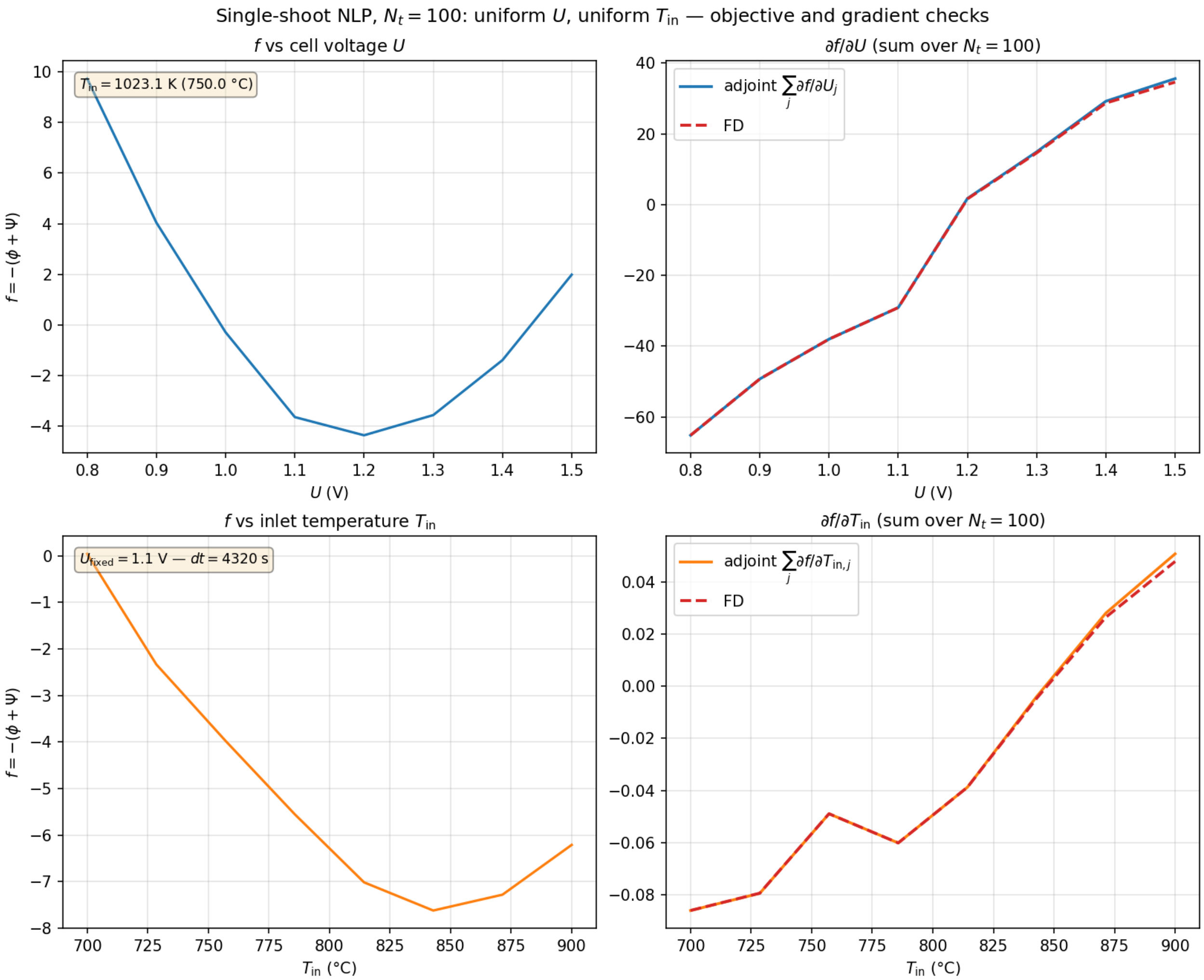


Figure A7.1: Single-shooting NLP gradient check at $N_t = 100$: objective $f = -(\phi + \psi)$ and its derivatives w.r.t. uniform cell voltage U (top) and inlet temperature T_{in} (bottom). Solid lines: adjoint; dashed: central FD with all segment controls perturbed together.

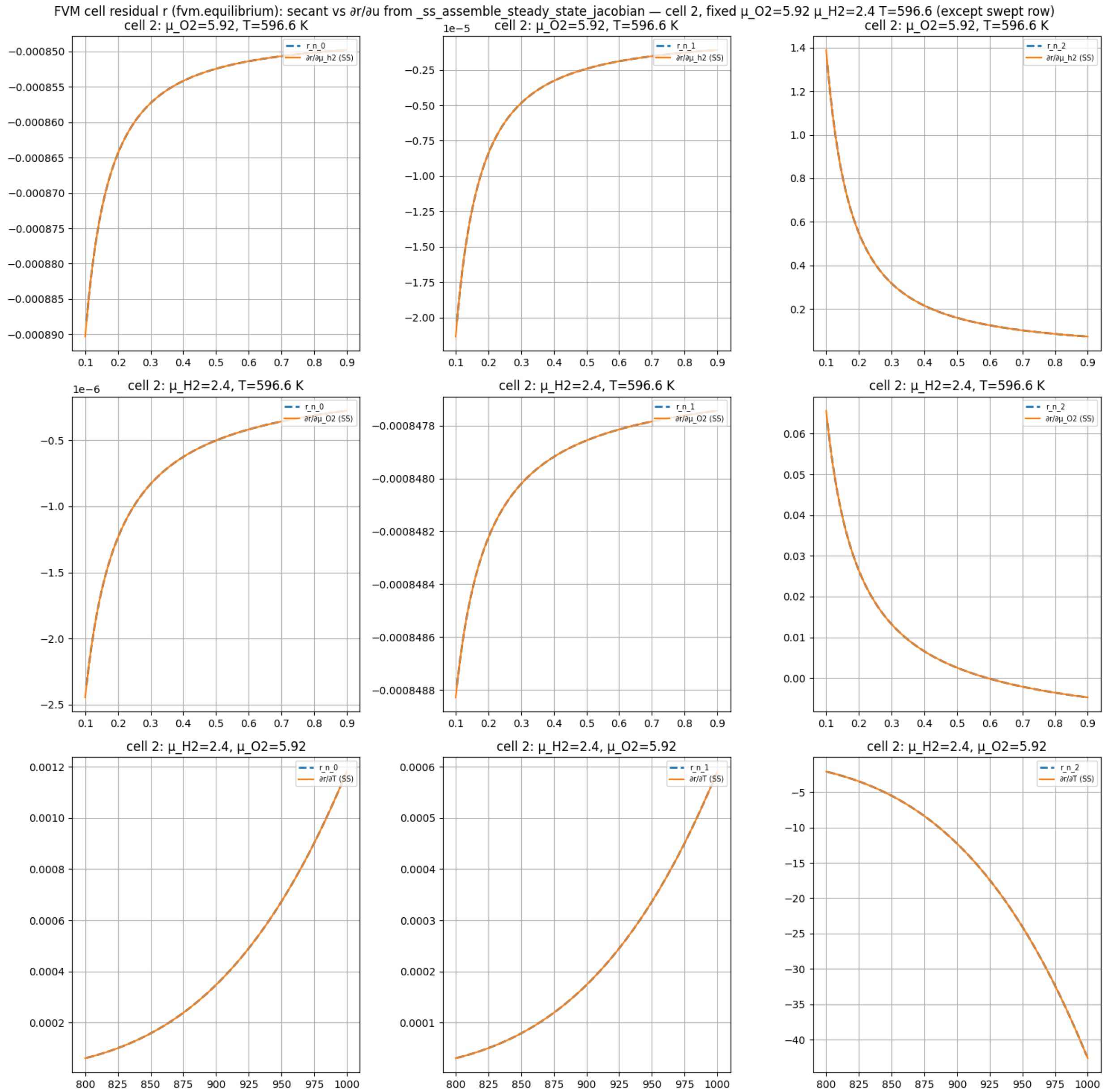


Figure A7.2: Steady-state FVM residual Jacobian vs. secant FD for the three residual components r_0, r_1, r_2 and inputs μ_{H_2}, μ_{O_2}, and T. Dashed: numerical secant; solid: analytical steady-state Jacobian.

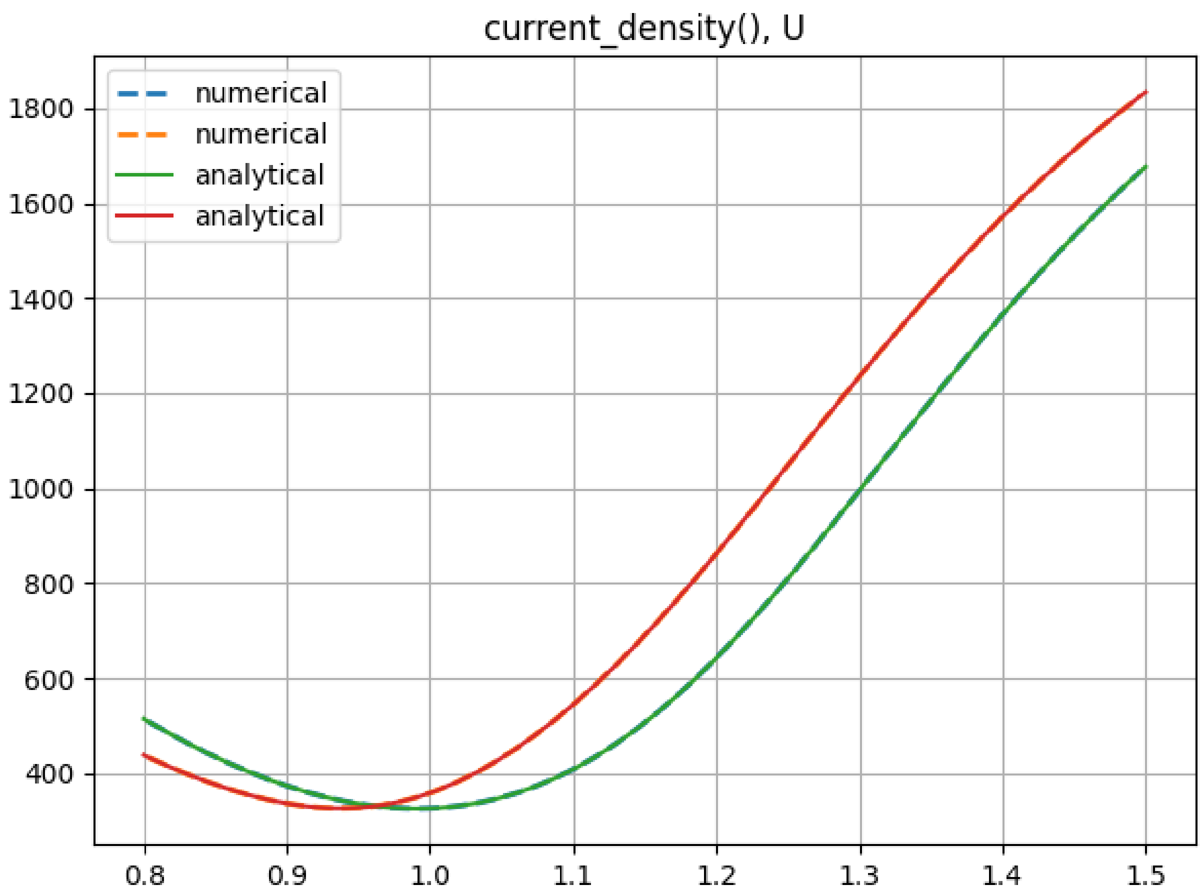


Figure A7.3: Analytical vs. numerical gradient of cell current density w.r.t. cell voltage U.

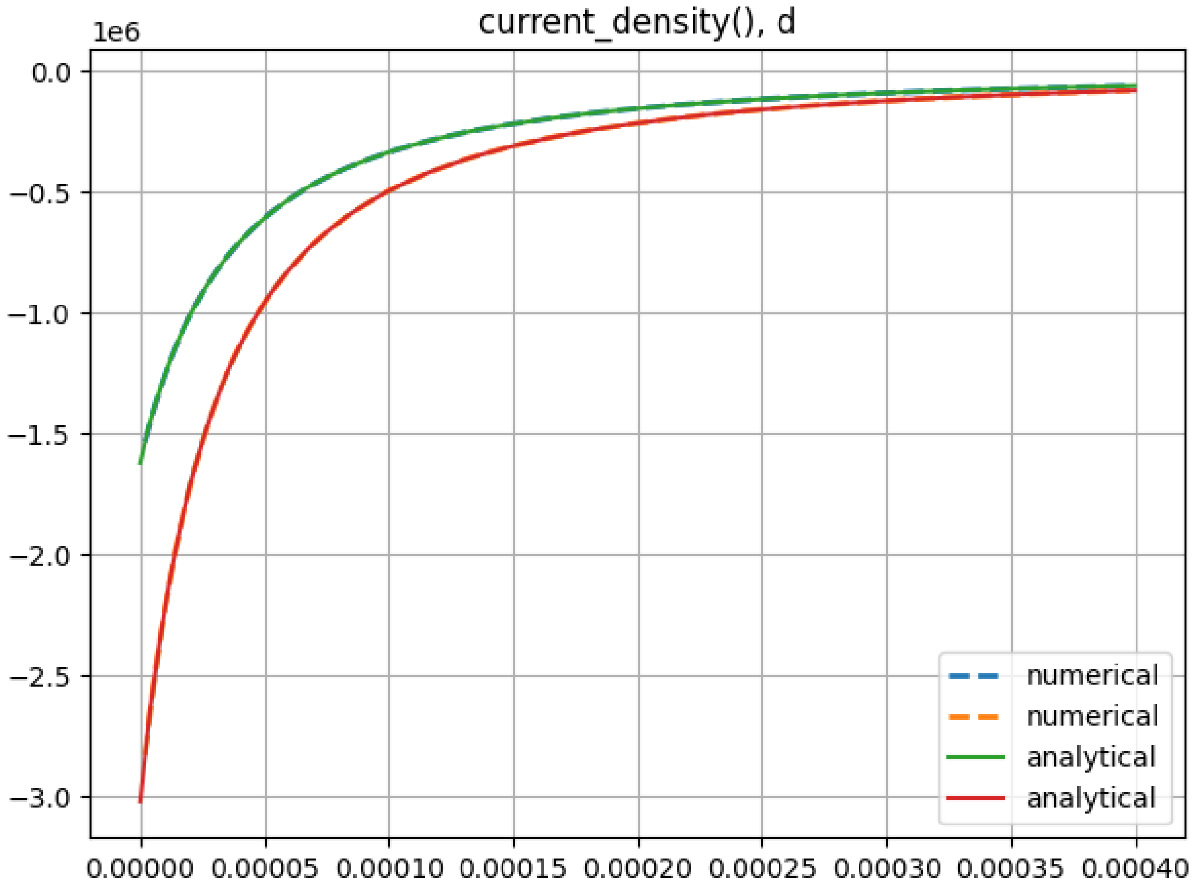


Figure A7.4: Analytical vs. numerical gradient of cell current density w.r.t. local degradation state d.

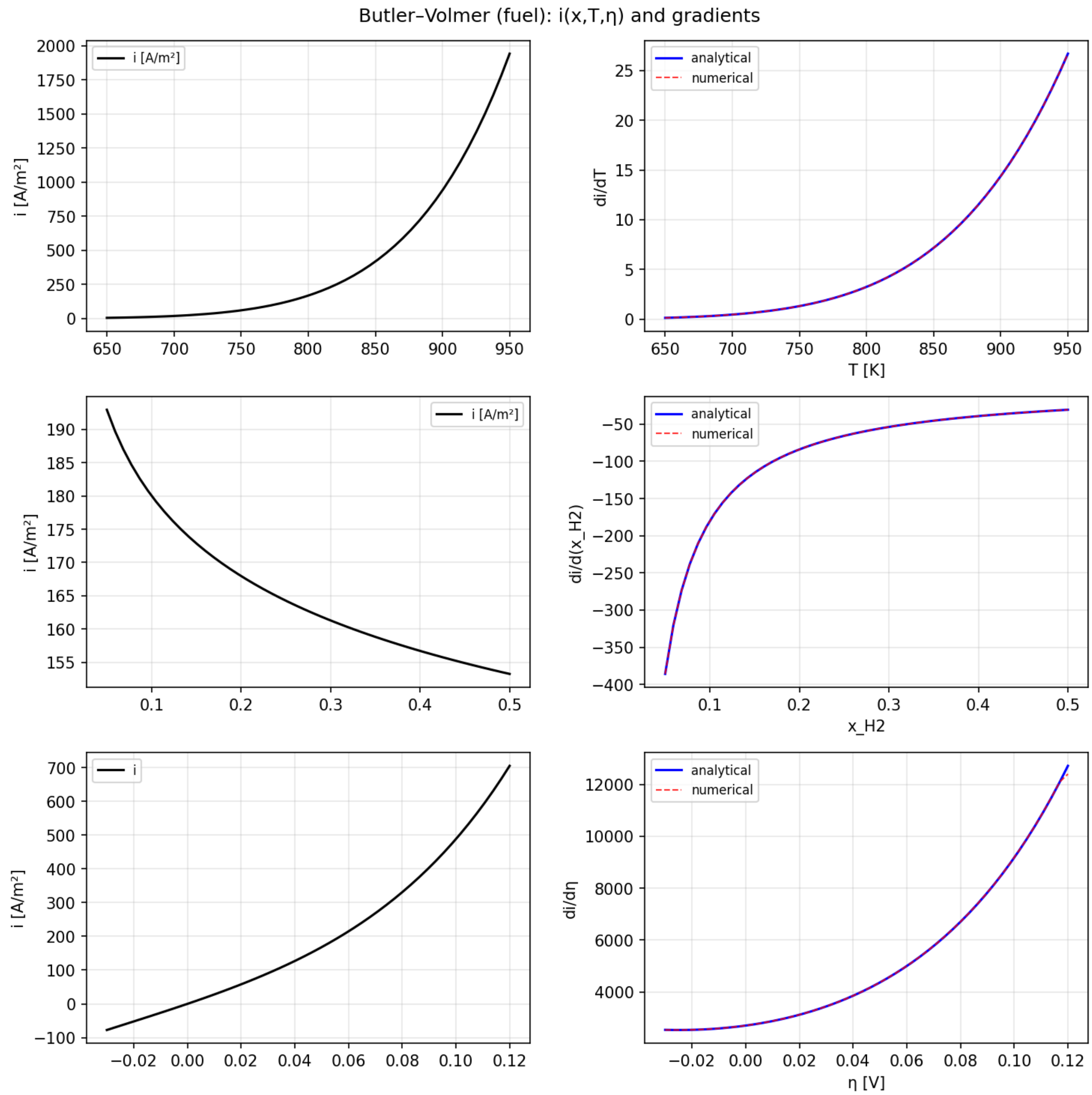


Figure A7.5: Fuel-side Butler–Volmer current $i(x, T, \eta)$ and gradients w.r.t. T, x_{H_2}, and activation overpotential η. Left column: i; right column: analytical (solid) vs. numerical (dashed) partial derivatives.

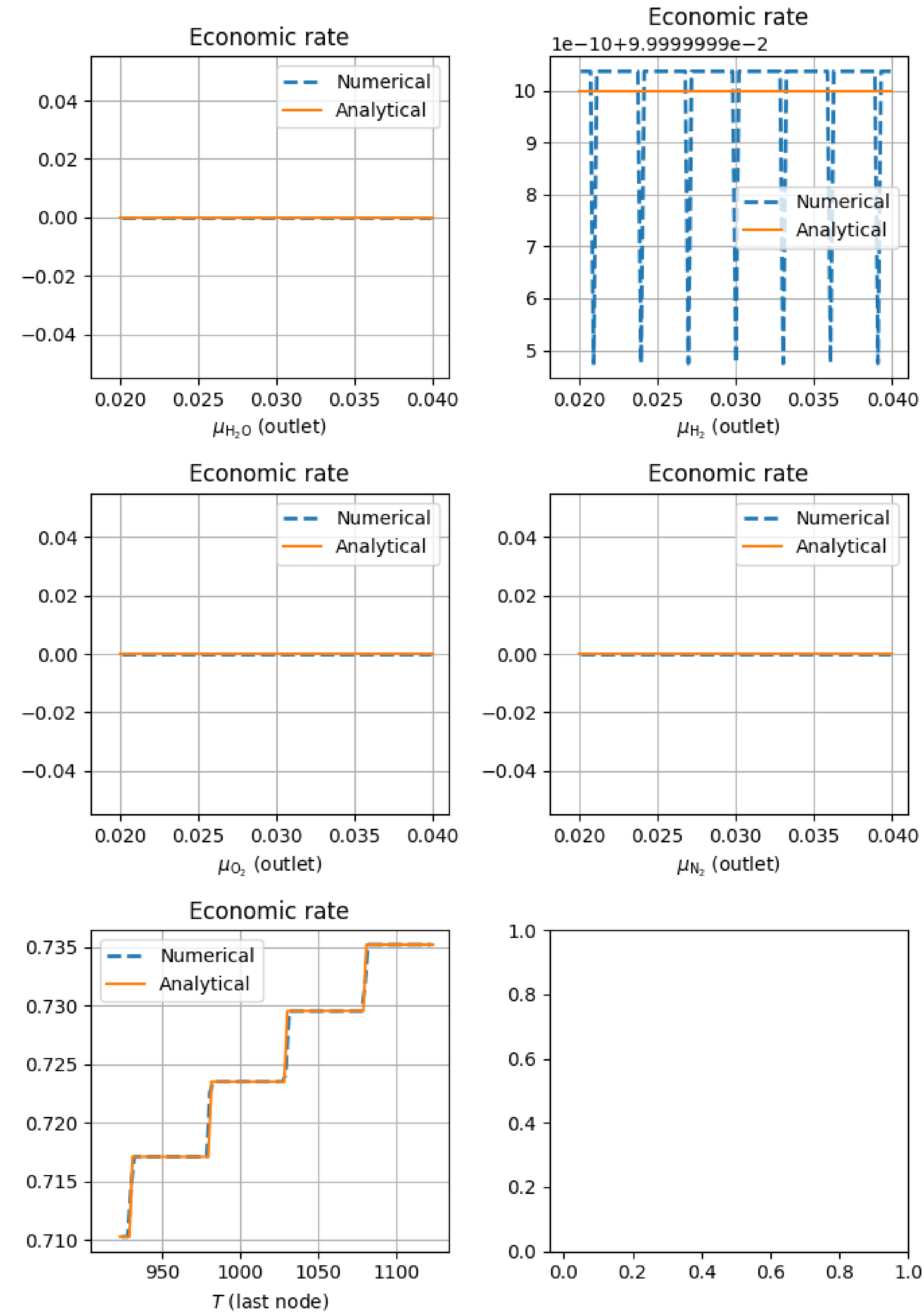


Figure A7.6: Analytical vs. numerical gradients of the instantaneous economic rate w.r.t. outlet species chemical potentials and the last-node temperature T.